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RESEARCH

Multi-objective UAV trajectory planning algorithm based on the hybrid MS-INSGA2 strategy

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Abstract

To address the critical challenges in 3D UAV trajectory planning—specifically the severe tendency to fall into local optima and the inherent dilemma of balancing convergence speed with population diversity in multi-objective optimization—this paper proposes a unified optimization framework integrating multiple mechanisms, namely the Multi-Strategy Improved Non-dominated Sorting Genetic Algorithm II (MS-INSGA2). This framework is founded on a linear interpolation mechanism with safety verification, effectively guaranteeing the safety of initial solutions and the quality of exploration starting points. During the evolutionary process, dynamic guidance and adaptive crossover and mutation mechanisms are deeply coupled, enabling real-time adaptive parameter adjustments based on the population status. This successfully breaks the deadlock between convergence and diversity. Subsequently, a local search strategy is introduced to meticulously refine the Pareto front, comprehensively enhancing trajectory smoothness and overall quality. Comprehensive evaluations on the ZDT, DTLZ, and WFG benchmark suites, supported by the Wilcoxon signed-rank test, confirm the superior underlying optimization capabilities of the proposed framework. Furthermore, in two complex 3D trajectory planning tasks, MS-INSGA2 demonstrates overwhelming superiority over 11 state-of-the-art algorithms, including MOBA, MOGOA, MOGWO, IMOABC, MOSSA, MOFA, MOEA-2DE, MPSOD, NSGA2, INSGA2, and DN-NSGA2. Specifically, it significantly reduces the Average Fitness by 27.87% and 32.06%, and decreases the Best Fitness by 6.91% and 14.18%, respectively. The research findings indicate that this unified framework not only substantially shortens the flight distance but also drastically mitigates the spatial collision risk, providing robust algorithmic support for highly reliable replanning in complex airspaces.

Keywords Hybrid MS-INSGA2 algorithm · UAV trajectory planning · Dynamic guidance · Adaptive crossover and mutation · Linear interpolation mechanism

1 Introduction

Unmanned aerial vehicle (UAV) trajectory planning is of great value in multi-scenario applications, but it also faces multiple stringent technical challenges. In urban low-altitude environments, existing planning methods still face challenges in balancing trajectory optimization and collision avoidance constraints (Zhang et al. 2021). In disaster rescue scenarios, conventional transportation modes struggle to fully adapt to the demands of complex terrain and high uncertainty (Jin et al. 2025). In emergency logistics, existing algorithms have room for further optimization in fast-response coordination (Zhang et al. 2026). Moreover, some traditional 2D methods are restricted when extended to 3D complex environments, and exclusive safe distance guarantees remain to be enriched (Li et al. 2018). Facing the test of balancing global search and fine-grained constraints for

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logistics UAVs, exploring novel trajectory planning schemes with strong environmental adaptability and multi-objective considerations is of great practical significance (Deng et al. 2025a, b). Meanwhile, cutting-edge methods such as reinforcement learning are still in the continuous exploration stage regarding sample efficiency and multi-UAV collaboration. Against this background, the NSGA-II (Non-dominated Sorting Genetic Algorithm II) algorithm, as a representative achievement of multi-objective evolutionary algorithms, can well balance the convergence and diversity of paths and has received extensive attention (Deb et al. 2002). However, in the face of complex UAV operational environments, the classical NSGA-II algorithm has gradually exposed certain theoretical limitations, restricting its efficacy in practical applications.

There is a direct restrictive relationship between the mechanistic defects of classical multi-objective algorithms and the practical application requirements of UAV trajectory planning. First, there is a lack of dynamic guidance strategies in balancing convergence and diversity; this deficiency in diversity preservation causes UAVs to easily fall into local dead ends in complex obstacle environments. Second, the crossover and mutation parameters of classical algorithms are typically fixed values; this parameter rigidification severely limits the dynamic adaptability of UAVs when shuttling between open airspace and dense threat zones. Furthermore, when dealing with complex high-dimensional problems such as non-convex and multi-modal landscapes, the efficiency of initial solution generation is low, making it difficult to meet the rapid response requirements of UAV emergency missions (Deng et al. 2025a, b). Finally, when facing optimization problems with stringent spatiotemporal and physical constraints, traditional mechanisms are relatively weak in local search capability, resulting in generated trajectories that are difficult to strictly satisfy physical limitations such as the maximum turning angle and smoothness of UAVs.

In response to the aforementioned challenges, a large number of fruitful improvement studies have emerged in recent years. To deeply analyze the evolutionary context of existing technologies, this paper systematically categorizes related literature into four major classes based on core optimization mechanisms, and focuses on discussing their advantages and limitations in UAV trajectory planning scenarios:

1. Initialization Strategies. Representative studies generate initial populations through task priority guidance (e.g., W-NSGAII (Guided Non-dominated Sorting Genetic Algorithm II) (Che et al. 2021)) or integrated K-means clustering (e.g., INSGA2 (Yeh 2019)). The advantage lies in significantly shortening the blind exploration time in the early stage of the algorithm and improving the feasibility of the initial solutions. The limitation is that these purely mathematical mechanisms based on spatial distribution lack a deep binding with the 3D physical environment and terrain of UAVs. When facing sudden radar threat zones, they are highly prone to generating invalid solutions that directly cross high-risk areas, which conversely increases the computational burden of subsequent evolution.
2. Adaptive Parameter Control. Such mechanisms balance global and local search by dynamically adjusting crossover and mutation operators (e.g., hybrid MMA-NSGAII (Mayfly Algorithm-Non-dominated Sorting Genetic Algorithm II) (Xiong et al. 2024) and scenario-based adaptation strategies (Zhaoyun et al. 2023)). The advantage is that they endow the algorithm with flexibility at different evolutionary stages. The limitation is that existing parameter adaptations mostly rely on the temporal dimension of "iteration number" and lack state perception of "population spatial crowding." When a UAV suddenly falls into a dense local obstacle matrix, the algorithm cannot achieve an instantaneous surge in the mutation rate to assist individuals in effective escape.
3. Local Exploitation. Researchers enhance local search by integrating sequential quadratic programming (NSGAII-CAM (Non-dominated Sorting Genetic Algorithm II-Goal Achievement Method) (Hong et al. 2022)), bottleneck-driven mechanisms (MNSGAII-BLS (Memetic NSGA-II algorithm with Bottleneck-driven Local Search) (Sun et al. 2026)), or simulated annealing (OSA (Suman et al. 2010)). The advantage is the ability to perform high-precision fine-tuning on initially generated trajectories. The limitation is that if local search is applied indiscriminately to the entire population, the computational overhead will increase exponentially, failing to meet the rapid response requirements of UAVs; furthermore, a lack of targeted optimization for elite individuals can easily result in the waste of onboard computational power.
4. Hybrid Multi-Objective Evolutionary Algorithms (Hybrid MOEAs). This category focuses on the design of global balancing mechanisms, such as introducing fuzzy logic (FL-NSGAII (Fuzzy Logic Guided Non-dominated Sorting Genetic Algorithm II) (Lau et al. 2009)), sequential solution generation (NSGAII-SSGS (Non-dominated Sorting Genetic Algorithm II-Sequential Solution Generation Strategy) (Zhang et al. 2022)), dynamic distribution adaptation (MODDA (Li et al. 2023)), and strategies combining two-stage co-evolution with reinforcement learning (Zhou et al. 2024). The advantage is the ability to obtain highly well-distributed and diverse non-dominated solution sets. The limitation is that such general evolutionary frameworks based on the objective space often decouple from specific UAV flight constraints (e.g., maximum turning angle, energy

consumption decay laws). As a result, solutions that perform excellently purely in mathematical benchmark tests (e.g., ANSGA-III (Jain and Deb 2014), RNSGA-II (Deb and Sundar 2006)) may face unflyable conditions or pose significant safety hazards in actual physical flight spaces.

The limitations exposed by the aforementioned mechanisms in UAV scenarios exactly constitute the direct motivation for proposing the multi-strategy integrated framework in this paper. The purely mathematical initialization and physics-decoupled hybrid evolution in categories 1 and 4 are highly prone to failure in actual terrain, which requires us to directly embed UAV safety verification and physical constraints into the underlying algorithm. On the other hand, the rigid parameter adaptation and expensive local search in categories 2 and 3 lack sensitivity and computational economy in complex collision avoidance, which prompts us to design crossover and mutation based on "diversity state feedback," as well as targeted local search "only for the elite Pareto front." Conventional discrete strategy stacking easily causes strategy redundancy and computational bottlenecks, making it difficult to substantively satisfy multiple stringent requirements simultaneously, such as optimal energy consumption and threat avoidance. Therefore, there is an urgent need for a comprehensive framework that can deeply couple various optimization components in underlying logic and highly integrate them with the physical topological features of UAVs.

To systematically bridge the gap between existing algorithms and practical application requirements, this paper proposes a multi-strategy synergistic UAV trajectory planning framework (MS-INSGA2). Rather than blindly piling up improvement strategies, this framework constructs deep synergistic effects closely aligned with the UAV flight operation logic of "environmental modeling—global optimization—local collision avoidance—trajectory smoothing." Its core contributions are specifically embodied in the coordination of the following dimensions:

1. Construction of a safety and physical feasibility guarantee layer: An initial solution generation mechanism based on endpoint linear interpolation is proposed, incorporating a simulated annealing local optimization strategy targeting the Pareto front. Specifically, a globally topologically safe search corridor for the UAV is constructed through interpolation initialization, and the Pareto front solutions are subsequently refined using the local optimization strategy. Relying on this closed-loop mechanism combining "macroscopic topological limitation and microscopic constraint refinement," it effectively compensates for the deficiencies of general adaptive NSGA-II variants in handling complex physical constraints.
2. Formulation of a global exploration and multi-objective balancing layer: A dynamic synergistic mechanism based on Pareto front mean guidance and state-dependent crossover and mutation is proposed. Within the safe domain, adaptive mutation dynamically adjusts the individual search step size based on population diversity to escape local optima; meanwhile, the front mean guidance acts as a global constraint, pulling the mutation trajectory steadily towards the overall optimal solution characterized by low energy consumption and high safety. This achieves a deep balance between complex penetration and energy consumption convergence in long-endurance UAV missions.
3. Multi-level performance verification and statistical analysis: To comprehensively evaluate the reliability of the framework, verification is conducted in UAV trajectory scenarios, and the ZDT, DTLZ, and WFG series of benchmark test functions are introduced to strictly evaluate its underlying optimization performance. Furthermore, the Wilcoxon signed-rank test is employed to analyze the experimental results, demonstrating the statistical significance of the algorithm's performance improvement.

The remainder of this paper is organized as follows: Section 2 outlines the workflow of the baseline algorithm (NSGA2). Section 3 introduces the proposed hybrid strategy MS-INSGA2 algorithm, detailing the integration of the dynamic guidance strategy, adaptive crossover and mutation strategy, linear interpolation mechanism, and fused local search. Section 4 utilizes the ZDT, DTLZ, and WFG benchmark functions to test and validate the MS-INSGA2 algorithm against 11 other algorithms, including MOBA, MOGOA, MOGWO, IMOABC, MOSSA, MOFA, MOEA-2DE, MPSOD, NSGA2, INSGA2, and DN-NSGA2. Section 5 establishes the multi-objective functions for the UAV flight environment and, based on this model, conducts comparative experiments in Task I and Task II between MS-INSGA2 and the 11 comparison algorithms. It further discusses the MS-INSGA2 algorithm using the Wilcoxon signed-rank test and the coefficient of variation (CV) of fitness. Section 6 conducts a sensitivity experimental analysis of the multiple parameters involved in the multi-strategy improvements of MS-INSGA2. Finally, Section 7 summarizes the entire study and provides suggestions for future research directions.

2 Basic nsga2 algorithm

Before delving into the proposed multi-strategy synergistic unmanned aerial vehicle (UAV) trajectory planning framework (MS-INSGA2), this section first reviews the core mechanisms of the baseline NSGA-II algorithm. Systematically outlining

baseline operations—such as non-dominated sorting, crowding distance calculation, genetic crossover and mutation, and elitism preservation—serves not only to construct the underlying theoretical framework of the proposed algorithm but, more importantly, to dissect the limitations exposed by traditional mechanisms when tackling complex 3D UAV trajectory planning tasks (e.g., the lack of environmental adaptability due to fixed crossover and mutation parameters, and the low early-stage exploration efficiency caused by the absence of high-quality initial solutions). This retrospective review provides a clear logical motivation and theoretical foundation for introducing the targeted improvement strategies (such as dynamic guidance and adaptive crossover and mutation) detailed in Sect. 3.

2.1 Non-dominated sorting

Randomly generate an initialization population $P = \{p_1, p_2, \cdots, p_n\}$, and set the population size to N. For each individual p_i in the population, the corresponding dominated individual is set to n_{pi}, and the dominated individual set is set to S_{pi}. The layer distribution in non-dominated sorting is shown in Fig. 1, which includes three fronts, denoted by F_i ($i \in \{1,2,3\}$). For any individuals P_i and P_j, if they satisfy Formula 1, it is denoted as $p_i \prec p_j$.

$$\begin{cases} f_k(p_i) \le f_k(p_j) \ \forall k \in \{1,2,3,\cdots,m\} \\ f_k(p_i) \le f_k(p_j) \ \exists\, k \in \{1,2,3,\cdots,m\} \end{cases} \tag{1}$$

2.2 Crowding degree calculation

To prevent individuals from becoming disproportionately concentrated in specific regions within the same non-dominated layer, the crowding degree is computed. The calculation involving three adjacent individuals in the crowding degree matrix is illustrated in Fig. 2, and the crowding distance is formulated as follows:

$$d_{p_i} = d_{p_i} + \frac{f_k(p_{i+1}) - f_k(p_{i-1})}{f_k^{\max} - f_k^{\min}} \tag{2}$$

where d_{p_i} denotes the crowding distance of individual p_i, and f_k represents the k-th objective function. The variables P_{i-1} and P_{i+1} refer to the preceding and succeeding adjacent individuals of P_i, respectively, after sorting based on the k-th objective function. Additionally, $f_k min$ and $f_k max$ denote the minimum and maximum values of the k-th objective within the current non-dominated layer.

2.3 Genetic crossover and mutation

The genetic crossover process preserves the high-quality features of the parent generation through inheritance while recombining new gene combinations. This facilitates the exploration and evolution of superior solutions within the solution space, effectively mitigating the risk of falling into local optima. The crossover mechanism is mathematically expressed as follows:

$$\begin{cases} p_i^{child_1} \leftarrow p_i^{child_1} \cup \{p_i^{gene_1} \cdot I(R < 0.5) + p_i^{gene_2} \cdot I(R \ge 0.5)\} \\ p_i^{child_2} \leftarrow p_i^{child_2} \cup \{p_i^{gene_2} \cdot I(R < 0.5) + p_i^{gene_1} \cdot I(R \ge 0.5)\} \end{cases} \tag{3}$$

where p_i^{child} and p_i^{child2} represent the gene sets of offspring 1 and offspring 2, respectively, while p_i^{gene1} and p_i^{gene2} denote the i-th gene of parent 1 and parent 2. The term $I(\cdot)$ stands for the indicator function,$\cup\{\cdot\}$ denotes the set addition operation, and $R \in (0,1)$ is a uniformly distributed random number. Subsequently, the mutation process introduces new genetic characteristics by applying random perturbations to the offspring generated by the crossover operation. This breaks fixed gene patterns, thereby preventing an excessive concentration of high-quality solutions in specific regions and ensuring the

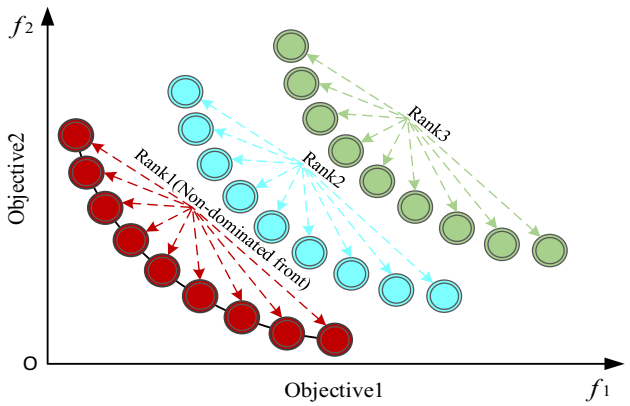


Fig. 1 Non-dominated sort each layer distribution

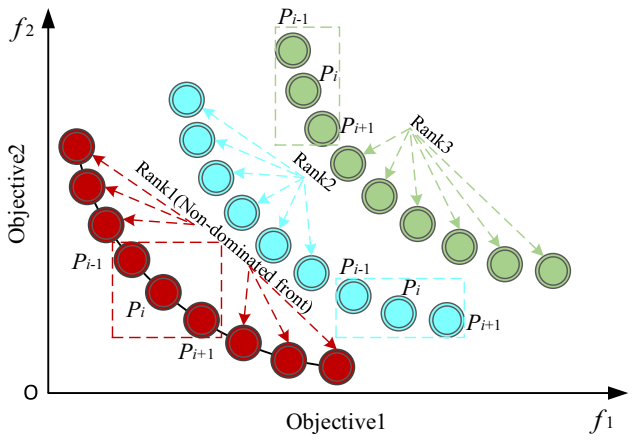


Fig. 2 Crowding degree matrix 3 adjacent calculation schematic diagram

uniformity and generalization capability of the solution set. The mutation step size factor δ is determined by:

$$\delta = \begin{cases} (2R)^{\frac{1}{\eta+1}} - 1 & R < 0.5 \\ 1 - [2(1-R)]^{\frac{1}{\eta+1}} & R > 0.5 \end{cases} \quad (4)$$

Here, η denotes the mutation distribution index, and R remains a random number within the range of (0, 1).

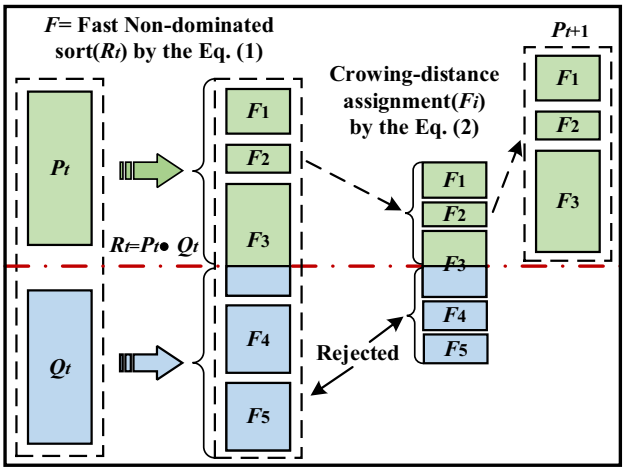


Fig. 3 Elite retention strategy graph

2.4 Elite retention strategy

During the non-dominated sorting process, to more effectively preserve superior individuals and eliminate inferior ones, an elitism preservation strategy is introduced, as illustrated in Fig. 3. The specific algorithmic procedure is detailed in Table 1.

3 Hybrid strategy MS-insga2 algorithm

To overcome the limitations of existing adaptive variants (e.g., ANSGA-III and RNSGA-II) that rely on preset reference points, as well as conventional hybrid frameworks that are prone to computational redundancy, this paper deeply couples UAV flight operational logic to construct a closed-loop collaborative architecture: preliminary safety delimitation, mid-term dynamic optimization, and terminal fine-grained smoothing. Specifically, the linear interpolation mechanism (Section 3.3) significantly compresses the invalid search space by pre-constructing a topologically safe spatial corridor. Building upon this, the dynamic guidance strategy (Section 3.1) utilizes the Pareto front mean as a guide to replace static reference points, driving the population to converge toward the optimal solutions. Simultaneously, the adaptive crossover and mutation strategy (Section 3.2)

Table 1 The elite retains the computational flow

Algorithm 1: Elitism Algorithm pseudocode

Input: Non-dominated frontier: F_i, Number of offspring: N, Parent population: P_t, Offspring population: Q_t

Output: Elite-preserved new population: P_{t+1}

$R_t = P_t \cup Q_t$

F = Fast Non-dominated-sort (R_t) by the Eq. (1)

$P_{t+1} = \varnothing$ and $i = 1$

until $|P_{t+1}| + |F_i| \le N$

  Crowding Distance Assignment (F_i) by the Eq. (2)

  $P_{t+1} = P_{t+1} \cup F_i$

  $i = i + 1$

Sort($F_i, \prec n$)

$P_{t+1} = P_{t+1} \cup F_i[1:(N - |P_{t+1}|)]$

Q_{t+1} = make-new-pop(P_{t+1})

$t = t + 1$

return P_{t+1}

dynamically adjusts the search step size based on the population status, assisting the algorithm in escaping local optima. Finally, the integrated local search mechanism (Section 3.4) executes simulated annealing optimization exclusively on the final Pareto front solutions, precisely refining constraints such as turning angles with minimal computational overhead. This deeply synergistic design avoids the mere stacking of strategies, ensuring computational efficiency while being highly adaptable to the stringent physical requirements of UAV trajectory planning. The mathematical models of these specific core mechanisms are detailed as follows:

3.1 Dynamic guidance strategy

The Pareto front mean is employed to guide individuals toward the optimal region. The front mean guidance model is formulated as follows:

$$\begin{cases} p\prime_i = p_i + s(\overline{p}_{front} - p_i) \\ s = 0.5 - \frac{0.4 \cdot t}{T_{\max}} \end{cases} \tag{5}$$

where $p\prime_i$ and p_i denote the chromosomes of the new individual after guidance and the original individual, respectively. The term $\overline{p}_{front}$ represents the mean value of the chromosomes of all individuals on the Pareto front. t indicates the current iteration number, T_{max} is the maximum iteration number, and s represents the guidance intensity, which decreases linearly from 0.5 to 0.1 as the iterations proceed. This achieves strong guidance for convergence in the early stage and weak guidance for diversity preservation in the later stage. The population diversity (denoted as D) is strictly defined as the standard deviation of the pairwise Euclidean distances between non-dominated individuals in the normalized objective space, which objectively reflects the dispersion of the solution set along the Pareto front. When the diversity D falls below a preset fixed empirical threshold of 0.15, the population is determined to be trapped in aggregation stagnation. At this point, the random perturbation model is triggered and added:

$$p_i\prime = p_i + \Delta p \Delta p \sim U(-200, 200) \tag{6}$$

The definitions of p'_i and p_i remain consistent with the above. Here, Δp denotes the perturbation amplitude drawn from a uniform distribution $U(-200, 200)$, which is used to apply large-scale perturbations to individual chromosomes and replace the individual with the lowest crowding degree in the population to enhance diversity.

3.2 Adaptive crossover and mutation strategy

To address the issue of fixed crossover and mutation rates in the NSGA2 algorithm, a dual-driven adaptive mechanism integrating time and state dependencies is proposed. This mechanism aims to balance the stability of macroscopic convergence and the sensitivity of local obstacle avoidance in trajectory planning.

① Time-dependent crossover scheduling mechanism: To provide a stable and predictable global convergence trend, a time-dependent model based on the number of iterations is introduced for the crossover rate ψ_c:

$$\psi_c = \psi_{c,init} - \frac{t}{T_{\max}}\left(\psi_{c,init} - \psi_{c,final}\right) \tag{7}$$

where ψ_c denotes the crossover probability at the current iteration t, and T_{max} indicates the maximum number of iterations. The parameters $\psi_{c,init}$ and $\psi_{c,final}$ represent the initial and final values of the crossover probability, respectively. This mechanism ensures that the algorithm maintains a relatively high crossover rate in the early stages of the search, thereby guaranteeing sufficient exploration of the global space.

② State-dependent mutation feedback mechanism: Unlike the pre-planned schedule of the crossover operation, mutation serves as an effective mechanism for escaping local optima. For the mutation rate ψ_m, a state-feedback model dependent on the population diversity D is introduced:

$$\psi_m = \psi_{m,\text{base}} + (1 - D) \cdot (\psi_{m,\max} - \psi_{m,\text{base}}) \tag{8}$$

where ψ_m represents the mutation probability at the current iteration, with $\psi_{m,base}$ and $\psi_{m,max}$ denoting its base and maximum values, respectively, and D represents the population diversity. Acting as a typical state-feedback driver, this mechanism can perceive the situation in real time when the UAV is trapped in dense local obstacle regions which causes a sudden drop in population diversity D and proportionally and rapidly increase the mutation rate to stimulate the escape capability of the individuals. Conversely, when the diversity is sufficient, the mutation rate is maintained at the base level.

To coordinate with the aforementioned operations, the mutation amplitude p_{range} similarly adopts a time-dependent smooth decay model. This prevents excessively large mutation step sizes in the later stages of iteration, which could cause the trajectory to collide with obstacles:

$$p_{\text{range}} = 80 - \frac{60 \cdot t}{T_{\max}} \tag{9}$$

Here, P_{range} defines the amplitude range of the mutation probability, while t and T_{max} retain their previous definitions. Through the organic combination of time-dependent and state-dependent strategies, the algorithm effectively enhances its adaptability in complex environments while sustaining population diversity.

3.3 Linear interpolation mechanism with safety verification

To address the issue that the completely random initial solutions generated by the classic NSGA2 algorithm are highly prone to producing a large number of invalid solutions crossing threat zones, which consequently leads to low exploration efficiency in the early stages of the algorithm, a linear interpolation initialization strategy with safety verification is proposed (detailed pseudocode is presented in Algorithm 2 within Table 2). First, linear interpolation is performed along the line connecting the starting point and the target point to generate base control points::

$$\begin{cases} x_i = x_{start} + \varsigma \cdot (x_{target} - x_{start}) \\ y_i = y_{start} + \varsigma \cdot (y_{target} - y_{start}) \\ z_i = z_{start} + \varsigma \cdot (z_{target} - z_{start}) \end{cases} \tag{10}$$

where x_i, y_i and z_i represent the coordinates of the i-th interpolated control point. The coordinates of the starting and target points are denoted by x_{start}, y_{start}, z_{start} and x_{target}, y_{target}, z_{target}, respectively, and $\varsigma \in (0,1)$ dictates the interpolation proportion. However, pure linear interpolation points are highly likely to fall directly into radar obstacle regions. To confront the safety issues of interpolated points, real-time safety verification and random escape perturbation adjustment are introduced. Specifically, the algorithm calculates the distance between the generated control point (x_i,y_i) and the centers of the known obstacle set in real time. If the point is detected to fall within a high-risk zone (i.e., the distance is less than $^{x}radius + 250$ m), safety adjustment is triggered, applying random perturbation increments Δx and Δy that follow a large-step uniform distribution $U(-200, 200)$:

$$\begin{cases} x_i' = x_i + \Delta x \\ y_i^i = yi + \Delta y \end{cases} \tag{11}$$

Table 2 Pseudocode procedure for linear interpolation with safety verification

Algorithm 2: Linear Interpolation Initialization with Safety Verification

Input: Start point: $(x_{\text{start}}, y_{\text{start}}, z_{\text{start}})$, Target point: $(x_{\text{target}}, y_{\text{target}}, z_{\text{target}})$, Number of control points: N_c, Set of radar obstacles: R, Map boundaries $[x_{\min}, x_{\max}], [y_{\min}, y_{\max}], [z_{\min}, z_{\max}]$

Output: A structurally safe chromosome C for a UAV trajectory

Initialize chromosome: $C \leftarrow \varnothing$

for i =1 to N_c **do**

  Compute interpolation proportion $\varsigma = i/(N_c + 1)$

  Generate base coordinates (x_i, y_i, z_i) via linear interpolation (Eq. 10)

  while point (x_i, y_i) is unsafe (Distance to any $r \in R \le r_{\text{radius}} + 250\text{m}$) **do**

    Generate random perturbation increments: $\nabla x \sim U(-300, 300)$, $\nabla y \sim U(-300, 300)$

    Apply perturbation and enforce boundary limits simultaneously: $x' = \max(x_{\min}, \min(x_i + \nabla x, x_{\max}))$, $y_i' = \max(y_{\min}, \min(y_i + \nabla y, y_{\max}))$ (Eq. 11-13)

    Update current point for next check: $x_i \leftarrow x_i', y_i \leftarrow y_i'$

  end While

  Bound altitude: $z_i = \max(z_{\min}, \min(z_i, z_{\max}))$

  Append the safe control point (x_i, y_i, z_i) to chromosome C

end for

Decode C into a continuous B-spline path and evaluate total Collision Risk

if Collision Risk > 5000 **then**

  return Algorithm 1 (Discard and regenerate recursively)

else

  return C

end if

where *x'i* and *y'i* represent the adjusted coordinates of the *i*-th control point, with Δx and Δy being the respective perturbation increments. To ensure that these large-step perturbations do not exceed the 3D physical airspace of the actual flight, strict boundary truncation constraints must be applied simultaneously:

$$x_i = x_{\text{start}} + \varsigma \cdot \Delta x_{\text{total}} \tag{12}$$

$$x\prime_i = \begin{cases} x_{\min} & \text{if } x_i + \Delta x < x_{\min} \\ x_i + \Delta x & \text{if } x_{\min} \le x_i + \Delta x \le x_{\max} \\ x_{\max} & \text{if } x_i + \Delta x > x_{\max} \end{cases} \tag{13}$$

Here, Δx_{total} represents the total spatial increment from the starting point to the target point along the *x*-axis, while x_{min} and x_{max} specify the allowable lower and upper limits of the *x*-coordinate, respectively. Through this threshold function, out-of-bound control points are forcefully pulled back into the feasible airspace. Through this threshold function, out-of-bound control points are forcefully pulled back into the feasible airspace.

Finally, after completing the local correction for all control points of a chromosome, the algorithm decodes it and evaluates the total collision risk (Collision Risk) of the global continuous path. If the total cost of the entire trajectory is abnormally high, the current local correction is deemed invalid; the algorithm will directly discard the individual and recursively trigger the regeneration mechanism (Algorithm 2), thereby thoroughly guaranteeing the physical feasibility and absolute safety of the initial population individuals at the root.

3.4 Integration of local search mechanism

To address the issue that the classic NSGA2 algorithm lacks fine-grained local exploitation of high-quality solutions after crossover and mutation, which easily leads to insufficient distribution precision of the Pareto front, this paper introduces a local search strategy based on Simulated Annealing (SA) at the end of each generation (i.e., after non-dominated sorting and environmental selection)(Ji et al. 2025). To balance the local exploitation capability and the global computational overhead, this strategy is not applied to the entire population. Instead, it strictly samples a 20% random subset of elite individuals from the current Pareto front to execute the local search.

For the selected individual $x_{current}$, its neighborhood solution x_{new} is generated by applying random perturbations following a uniform distribution to the coordinates of each trajectory control point in the 3D space. Specifically, the step size is $U(-50, 50)$, aiming to explore potentially superior paths near this high-quality solution. The solution acceptance mechanism of the local search follows the Metropolis criterion, which is formulated as follows:

$$x_{current} = \begin{cases} x_{new}, \Delta f < 0 \\ x_{new}, \Delta f \ge 0 \wedge rand(0,1) < exp\left(-\frac{\Delta f}{T}\right) \\ x_{current}, others \end{cases} \tag{14}$$

where Δf is the difference between the new cost value $f(x_{new})$ and the current cost $f(x_{current})$, and *T* represents the current temperature. This criterion allows the algorithm to accept inferior solutions with a certain probability, thereby effectively avoiding entrapment in local optima. The cooling process of simulated annealing is as follows:

$$T_{k+1} = T_k \times \alpha \tag{15}$$

In this cooling process, Tk and T_{k+1} denote the temperatures at the *k*-th and $(k-1)$-th steps, respectively, governed by the cooling coefficient α. To reasonably control the computational budget, the maximum number of SA iterations for each selected individual is set to 20, with the initial temperature set to $T = 100$ and the cooling coefficient $\alpha = 0.9$.

3.5 Overall execution flow of MS-insga2

The procedure of the hybrid MS-INSGA2 strategy, formulated by incorporating the dynamic guidance strategy, adaptive crossover and mutation strategy, linear interpolation mechanism, and integrated local search into the classic NSGA2 algorithm, is illustrated in Fig. 4 and Table 3. The specific steps are detailed as follows:

Step1: Initialize the population solution according to formula (10) - (13).
Step2: Perform fast non-dominated sorting on the individuals in the population to divide the Pareto front F_1,F_2,… And each individual is assigned a frontier number.
Step3: Calculate crowding distance for individuals in each Pareto front according to Formula (2).
Step4: According to Formula (5) and (6), dynamic guidance strategy is used to maintain population diversity and convergence.
Step5: Use binary tournament selection to generate mating pool and complete population selection operation.
Step6: According to Formula (7), the adaptive mutation strategy is used to realize the population mutation operation.
Step7: Merge parent and offspring populations according to Formula (14), and retain the elite retention strategy through non-dominated sorting and crowding degree screening.
Step8: Reach the maximum number of iterations, output the final Pareto front solution set, otherwise return to step 2 and repeat the iteration.

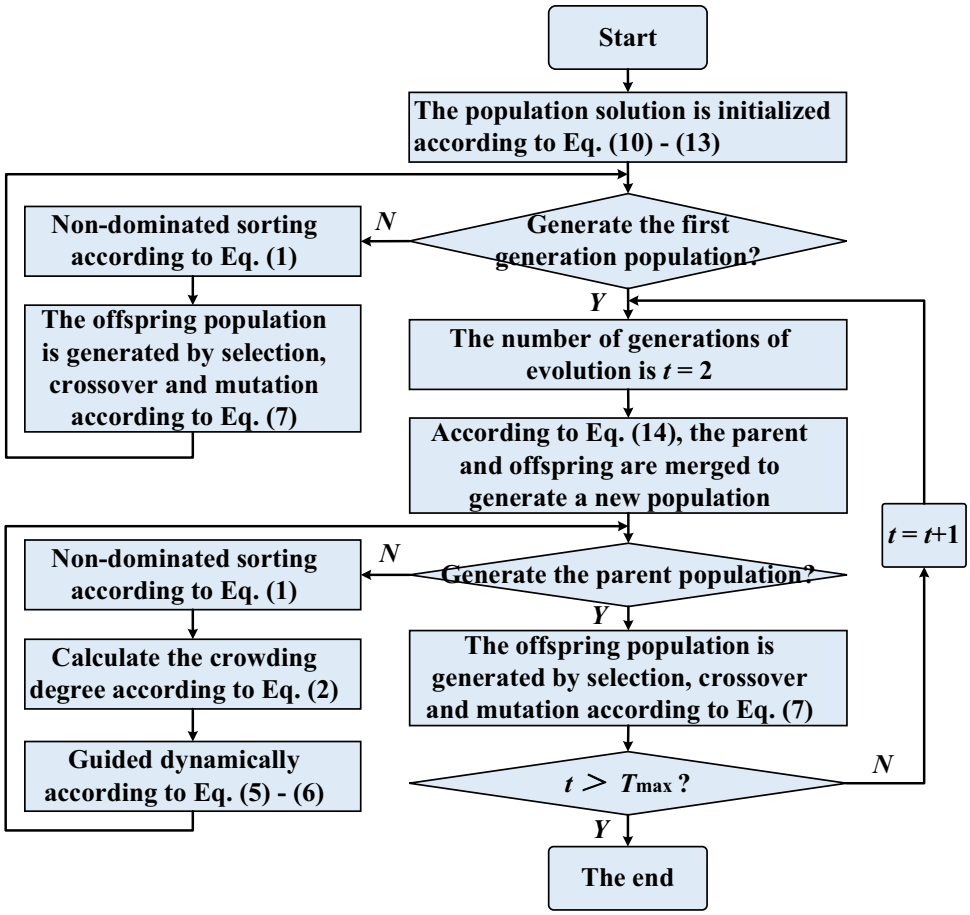


Fig. 4 Flowchart of the MS-INSGA2 hybrid strategy

4 Basic model testing

4.1 Experimental settings

In To verify the effectiveness of the improved MS-INSGA2 hybrid strategy algorithm, this paper utilizes the ZDT, DTLZ, and WFG benchmark test suites and introduces 11 mainstream and classic algorithms for comparative validation: MOBA, MOGOA, MOGWO, IMOABC, MOSSA, MOFA, MOEA-2DE, MPSOD, NSGA2, INSGA2, and DN-NSGA2. In the experiments, the population size for all 12 algorithms is uniformly set to $N=50$, and the number of iterations is set to $M=100$. The internal parameters of each comparison algorithm are configured with reference to their original literature, and the detailed parameter settings for all evaluated algorithms are summarized in Table 4.

4.2 Evaluation indicators of the basic model

4.2.1 Inverted generational distance

The Inverted Generational Distance (IGD) (Zuo et al. 2025; Hu et al. 2024) is a comprehensive metric used to evaluate the convergence and diversity of the obtained solutions. It is computed by calculating the average shortest distance from the points on the true Pareto front to the generated solution set, where a smaller IGD value indicates superior algorithmic performance. The specific calculation is formulated as follows:

$$IGD(P,S)=\frac{\sqrt{\sum_{i=1}^{|p|}d_i^2}}{|P|} \tag{16}$$

$$d_i=\min_{x\in S}\|F(p^i)-F(x)\|\ p^i\in P \tag{17}$$

where P denotes the sampling set of the true Pareto front, and S represents the approximate Pareto set obtained by the evaluated algorithm. The term d_i represents the minimum Euclidean distance from the i-th point p^i in P to any point x in S. Furthermore, $F(p^i)$ and $F(x)$ denote their corresponding objective function value vectors in the objective space, and $|P|$ indicates the cardinality of the set P.

4.2.2 Hausdorff distance

The Hausdorff Distance (HD) (Santos and Xavier 2018; Schutze et al. 2012) is utilized to assess the proximity between solution sets, and its calculation is mathematically formulated as follows:

Table 3 Pseudocode procedure of the MS-INSGA2 algorithm

Algorithm 3: MS-INSGA2 Framework

Input: Start point $(x_{\text{start}}, y_{\text{start}}, z_{\text{start}})$, Target point $(x_{\text{target}}, y_{\text{target}}, z_{\text{target}})$, Population size N, Max generation $T_{\max}$

Output: The optimal Pareto front trajectory set F_1

$P_0 \leftarrow \varnothing$

for $i = 1$ **to** N **do**

  Generate the i-th control point coordinates x_i, y_i, z_i according to the interpolation ratio $\varsigma \in (0,1)$ (Eq. 10)

  Introduce perturbation increments $(\nabla x, \nabla y)$ and perform safety checks with boundary limits $[x_{\min}, x_{\max}]$ (Eq. 11-13)

  Add the generated safe individual p_i to the initial population: $P_0 \leftarrow P_0 \cup \{p_i\}$

end for

Perform fast non-dominated sorting on P_0 (Eq. 1) to divide it into fronts F_1、F_2,..., and calculate the crowding distance d_{p_i} (Eq. 2)

Initialize current generation $t \leftarrow 1$

while $t \le T_{\max}$ **do**

  Extract the Pareto front F_1 of the current population P_t, and calculate the mean of all chromosomes in the front $\overline{p}_{\text{front}}$

  for each individual $p_i \in P_t$ **do**

    Calculate the guidance strength $s = 0.5 - \dfrac{0.4 \cdot t}{T_{\max}}$, and guide the individual towards the optimal region: $p_i' = p_i + s(\overline{p}_{\text{front}} - p_i)$ (Eq. 5)

    If population diversity is below the threshold, add a random large-scale perturbation amplitude: $p_i' = p_i + \Delta p$ (Eq. 6)

  end for

  Generate the mating pool Ω_{MP} using tournament selection, and initialize offspring population $Q_t \leftarrow \varnothing$

  Evaluate the current population diversity D

  Calculate the adaptive crossover probability ψ_c based on current generation t (Eq. 7)

  Calculate the mutation probability ψ_m based on diversity D (Eq. 8), and calculate the mutation amplitude range p_{range} based on t (Eq. 9)

  for every two parent individuals sequentially drawn from Ω_{MP} **do**

    Perform genetic crossover based on probability ψ_c to generate offspring

    Apply random mutation perturbation to the generated offspring based on probability ψ_m and amplitude p_{range}

    Add the generated offspring into Q_t

  end for

  Merge parent and offspring populations to form $R_t \leftarrow P_t \cup Q_t$

  Select N elite individuals from R_t via fast non-dominated sorting (Eq. 1) and crowding distance assignment (Eq. 2) to generate P_{t+1}

  Extract the Pareto front F_1 of the new population P_{t+1}

  for each elite individual $x_{\text{current}} \in F_1$ **do**

    Generate a new local neighborhood solution x_{new} and calculate the objective difference $\nabla f = f(x_{\text{new}}) - f(x_{\text{current}})$

    Retain the new solution based on the probability of the Metropolis acceptance criterion (Eq. 14)

    Update the temperature according to the cooling coefficient $\alpha : T_k = T_{k-1} \times \alpha$ (Eq. 15)

  end for

  $t \leftarrow t + 1$

end while

return Perform non-dominated sorting on the final population $P_{T_{\max}}$ and output the optimal Pareto front set F_1

Table 4 Settings of various algorithm parameters

References	Algorithm	Parameter index	values
(Xu et al. 2024)	MOBA	α^{moba},γ^{moba},Q^{moba},A_0^{moba},r_0^{moba}	0.9, 0.9, [0,2], [0,2], [0,1]
(Mirjalili et al. 2018)	MOGOA	$c_{\max}^{mogoa}$,$c_{\min}^{\text{mogoa}}$,$f^{\text{mogoa}}l^{\text{mogoa}}$,	1, 0.00001, 0.5, 1.5
(Mirjalili et al. 2016)	MOGWO	a^{mogwo}, C^{mogwo}	[2,0], [0,2]
(Cui et al., 2023)	IMOABC	$\text{SN}^{\text{imoabc}}$,$Limit^{imoabc}$	100, 100
(Li and Wang 2022)	MOSSA	k,ST^{mossa},PE^{mossa},PD^{mossa}	2, [0.7,0.9], [0.2,0.3], [0.2,0.9]
(Zhao et al. 2023)	MOFA	α^{mofa},β_0^{mofa},γ^{mofa},	0.2, 1, 1
(Xu et al. 2024)	MOEA-2DE	$\eta_c^{\text{moea-2de}}$,$\eta_m^{\text{moea-2de}}$,$\text{CR}^{\text{moea-2de}}$,$\text{F}^{\text{moea-2de}}$,$\text{SD}^{\text{moea-2de}}$	0.9, 0.1, 0.3, 0.3, 0.3
(Pang et al. 2024)	MPSOD	ω^{mpsod},c_1^{mpsod},c_2^{mpsod}	[0.1,0.5],[1.5,2.5], [1.5,2.5]
(Guo et al. 2020)	NSGA2	c^{nsga2},m^{nsga2}	0.8, 0.8
(Duan et al. 2024)	INSGA2	C_p^{insga2},M_p^{insga2},L_m^{insga2}	0.8, 0.3, 20
(Lv et al., 2026, Liang et al. 2016)	DN-NSGA2	$p_c^{\text{dn-nsga2}}$,$\eta_c^{\text{dn-nsga2}}$,$p_m^{\text{dn-nsga2}}$,$\eta_m^{\text{dn-nsga2}}$	1, 20, 1/n, 20
This paper	MS-INSGA2	$\psi_{c,\text{init}}$,$\psi_{c,\text{final}}$,$\psi_{m,\text{base}}$,$\psi_{m,\max}$,α,$p_c^{\text{ms-insga2}}$,$p_m^{\text{ms-insga2}}$,$L^{\text{ms-insga2}}$	0.75, 0.65, 0.25,0.3, 0.9, [0.75,0.85], [0.15,0.25], 0.2

$$HD(X,Y) = max\left(\left(\frac{1}{N}\sum_{i=1}^{N} d_{dist}(x_i,Y)^{\varepsilon}\right)^{\frac{1}{\varepsilon}}, \left(\frac{1}{M}\sum_{i=1}^{M} d_{dist}(y_i,X)^{\varepsilon}\right)^{\frac{1}{\varepsilon}}\right) \tag{18}$$

where $X = \{x_1, x_2, \cdots, x_N\}$ and $Y = \{y_1, y_2, \cdots, y_N\}$ denote the sets of solutions evaluated on the True Pareto Front (TPF). The parameters N and M indicate the respective number of solutions in sets X and Y (both of which are subsets of the TPF). The function $d_{dist}(\cdot)$ computes the Euclidean distance, and ε serves as the weight parameter for adjusting the distance metric.

4.2.3 Standard deviation of minimum distance

The Spacing metric, defined as the standard deviation of the minimum distance among solutions (Bai et al. 2024), is utilized to evaluate the distribution uniformity of the solution set. A smaller Spacing value indicates superior uniformity and diversity. This metric is mathematically defined as follows:

$$Spacing = \sqrt{\frac{1}{|S|}\sum_{i=1}^{|S|}\left(d_i - \overline{d}\right)^2} \tag{19}$$

where S and d_i retain their previously established definitions, and $\overline{d}$ represents the average of all d_i values, reflecting the mean minimum distance between adjacent solutions within the set.

4.3 Basic test experiment

4.3.1 Testing functions

The test functions including ZDT, DTLZ and WFG functions are shown in Table 5 (Reddy and Dulikravich 2022). The ZDT function is a 2D test environment, and DTLZ and WFG are 3D test environments.

4.3.2 Experimental results and analysis

The mixed strategy MS-INSGA2 algorithm compares 11 algorithms such as MOBA, MOGOA, MOGWO, IMOABC, MOSSA, MOFA, MOEA-2DE, MPSOD, NSGA2, INSGA2 and DN-NSGA2 applied in ZDT and DTLZ Figs 5, 6, 7, 8, 9, and 10 show the visualizations of different test functions and WFG functions.

As listed in Table 5, the ZDT, DTLZ, and WFG series benchmark functions with distinct characteristics are selected for testing in this section. The quantitative statistical results (the mean and standard deviation of the IGD, HD, and Spacing metrics) of each comparison algorithm on these benchmark functions are comprehensively summarized in Tables 6, 7, 8, 9, 10, and 11, respectively. To visually demonstrate the algorithmic performance, the bold values in the tables indicate the optimal results achieved among all comparison algorithms. Furthermore, the results of the Wilcoxon rank-sum test at a 5% significance level are introduced at the bottom of the tables, denoted as W +/−/≈. Specifically, "+", "−", and "≈" indicate that the performance of the comparison algorithm is significantly superior to, significantly inferior to, or exhibits no significant difference from the proposed MS-INSGA2 algorithm, respectively.

As shown in Tables 6 and 7, MS-INSGA2 exhibits an overwhelming advantage on the ZDT benchmark suite, which evaluates bi-objective optimization capabilities. Regarding the comprehensive metrics IGD and HD that measure convergence and distribution, MS-INSGA2 secures the vast majority of the optimal values. According to the

Table 5 Benchmark test function

Type	Serial Number	Properties	Number of variables	Number of objectives
ZDT functions	ZDT1	Convex, Continues PF	30	2
	ZDT2	Concave, Continues PF	30	2
	ZDT3	Discontinuous, Disconnected PF	30	2
	ZDT4	Multi-modal, Convex, Continues PF	10	2
	ZDT6	Multi-modal, Concave, Non-uniform	10	2
DTLZ functions	DTLZ1	Multi-modal, Continues PF	7	3
	DTLZ2	Concave, Continues PF	12	3
	DTLZ3	Multi-modal, Concave	12	3
	DTLZ4	Concave, Highly non-uniform PF	12	3
	DTLZ5	Mostly degenerated, Concave	12	3
	DTLZ6	Mostly degenerated, Concave, Biased	12	3
	DTLZ7	Multi-modal, Disconnected PF, Mixed	22	3
WFG Functions	WFG1	Mixed, Biased	12	3
	WFG2	Multi-modal, Convex, Disconnected	12	3
	WFG3	Mostly degenerated, Non-separable	12	3
	WFG4	Multi-modal, Concave	12	3
	WFG5	Deceptive, Concave	12	3
	WFG6	Concave, Non-separable	12	3
	WFG7	Concave, Biased	12	3
	WFG8	Concave, Biased, Non-separable	12	3
	WFG9	Multi-modal, Concave, Deceptive, Biased	12	3

quantitative data from the Wilcoxon rank-sum test, the IGD value of the classic NSGA2 algorithm on the non-convex function ZDT2 is 7.3468E − 02, whereas MS-INSGA2 drastically reduces it by two orders of magnitude, reaching 3.0700E − 04. Across all five ZDT test functions, the comprehensive performance of most comparison algorithms, including the classic NSGA2, is significantly inferior to that of the improved MS-INSGA2. This demonstrates that the dynamic Pareto front guidance strategy introduced in this study effectively enhances the convergence accuracy of the algorithm in approaching the true Pareto front.

Tables 8 and 9 present the performance of the evaluated algorithms on the DTLZ benchmark suite, which is characterized by strong local convergence traps and high-dimensional features. Confronted with the conflicts inherent in high-dimensional objective spaces, MS-INSGA2 consistently maintains an exceptionally high quality of the solution set. Particularly on highly challenging functions such as DTLZ1, DTLZ2, and DTLZ6, both its IGD and Spacing metrics outperform those of the vast majority of meta-heuristic multi-objective algorithms (e.g., MOBA and MOGWO). For instance, on the DTLZ6 function, the IGD of MS-INSGA2 achieves an exceptionally low error level of 2.0700E − 04. The overwhelming number of negative (" − ") statistical results indicates that the adaptive crossover and mutation mechanism of MS-INSGA2 effectively overcomes the issue of diversity loss in high-dimensional spaces, thereby guaranteeing the stability of multi-objective balancing.

As demonstrated in Tables 10 and 11, the WFG benchmark suite possesses highly deceptive, degenerate, and non-separable features, posing an ultimate test to the underlying optimization robustness of the algorithms. The empirical data indicate that even within the complex WFG landscapes, MS-INSGA2 not only maintains its leadership in comprehensive metrics such as IGD and HD, but also performs exceptionally well in the Spacing metric, which reflects the uniformity of the solution set. Taking WFG2 and WFG3 as examples, MS-INSGA2 achieves the optimal performance across almost all metrics. This outstanding performance is credited to the integrated simulated annealing fine-grained local search strategy coupled at the terminal stage of the proposed framework. This mechanism enables the algorithm to precisely escape local optima with minimal computational overhead when confronting highly deceptive and complex objective spaces.

In summary, the quantitative data and the significance results from the Wilcoxon signed-rank test strongly corroborate the substantial performance superiority of MS-INSGA2 over the classical NSGA-II and 11 other mainstream multi-objective algorithms. To move beyond the superficial mechanistic description of merely "balancing convergence and exploration," this paper further conducts an in-depth analysis of the experimental results from three dimensions: problem

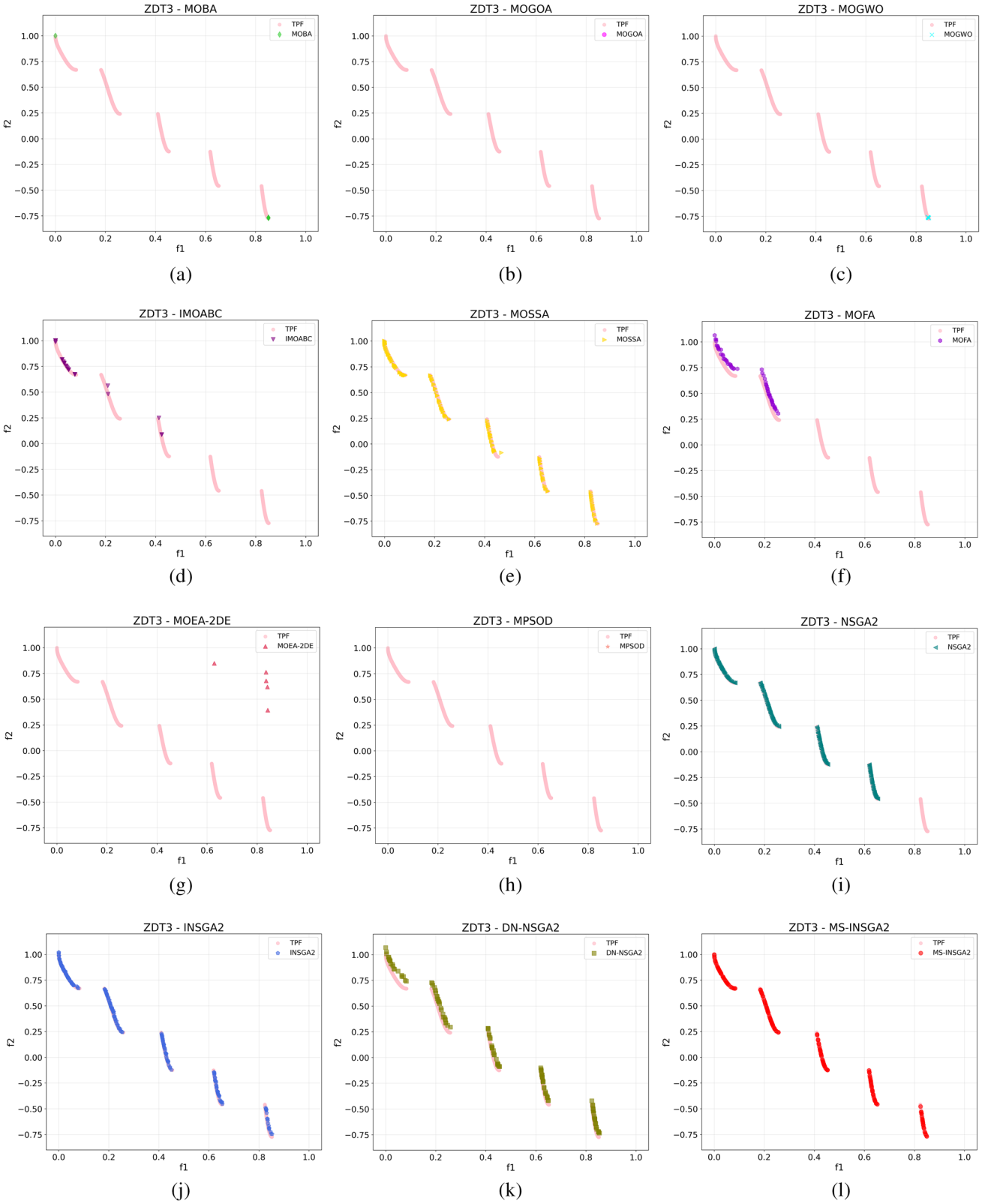


Fig. 5 ZDT3 represent (*a-l*)

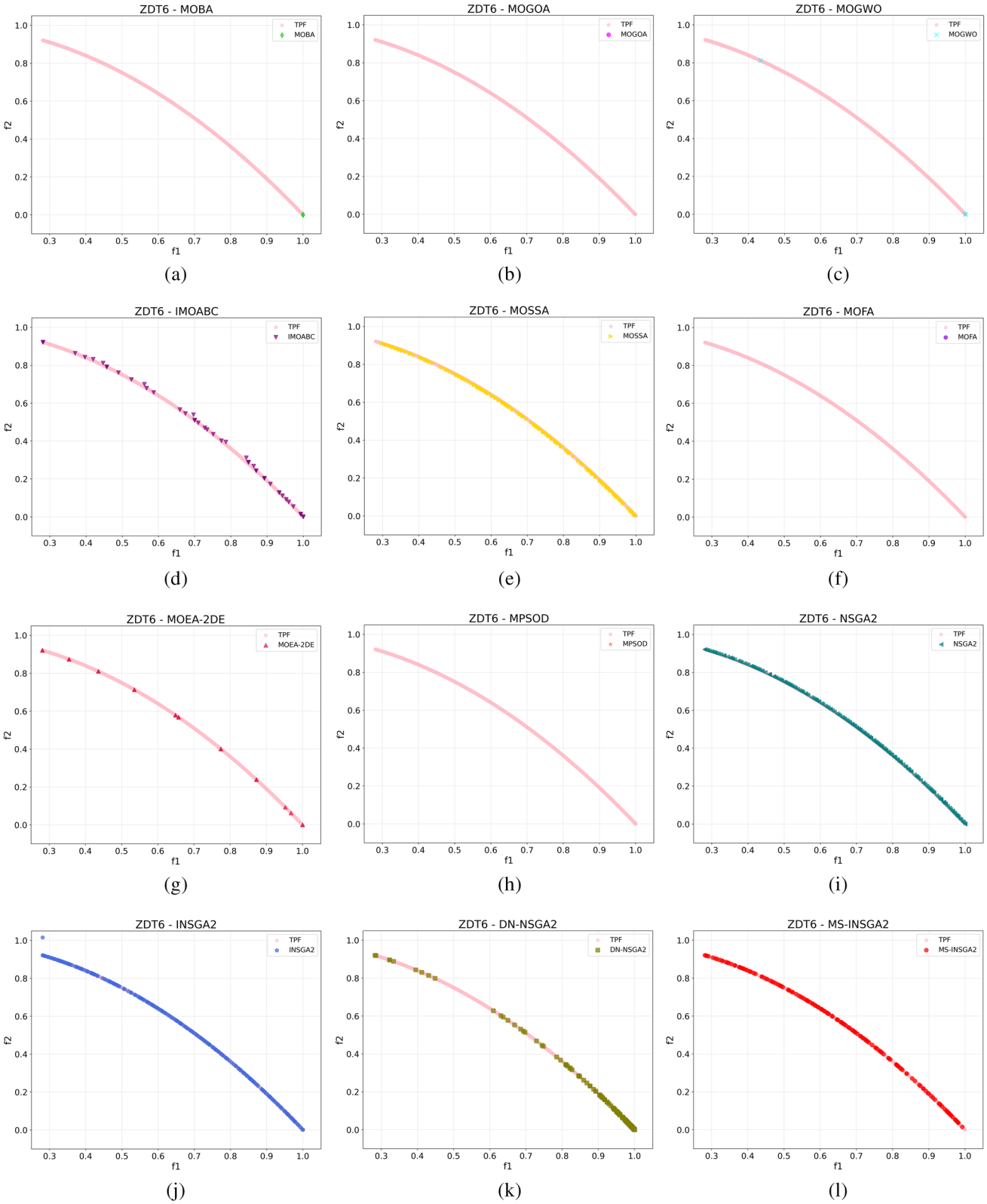


Fig. 6 ZDT6 represent (*a-l*)

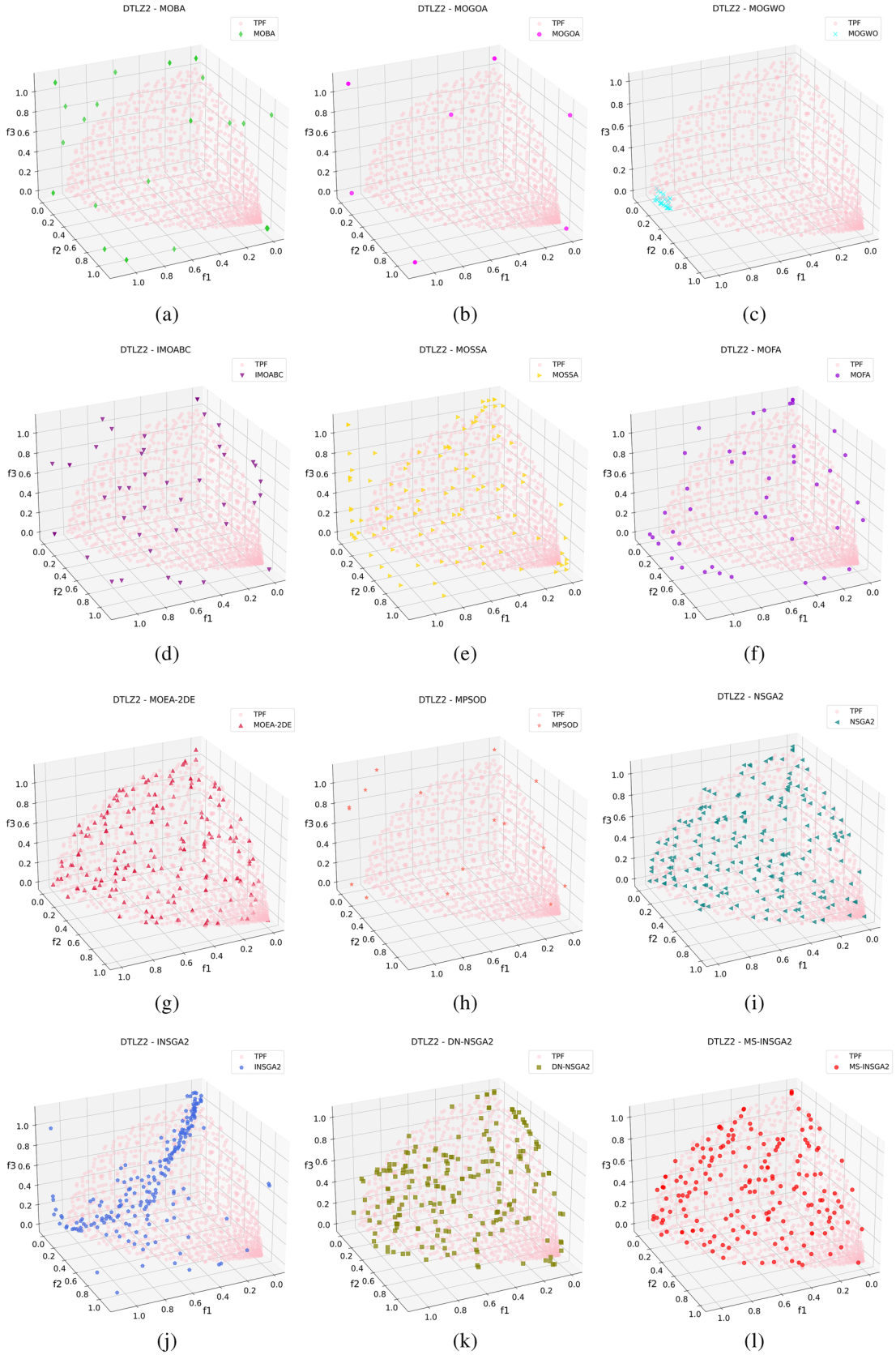


Fig. 7 DTLZ2 represent (*a-l*)

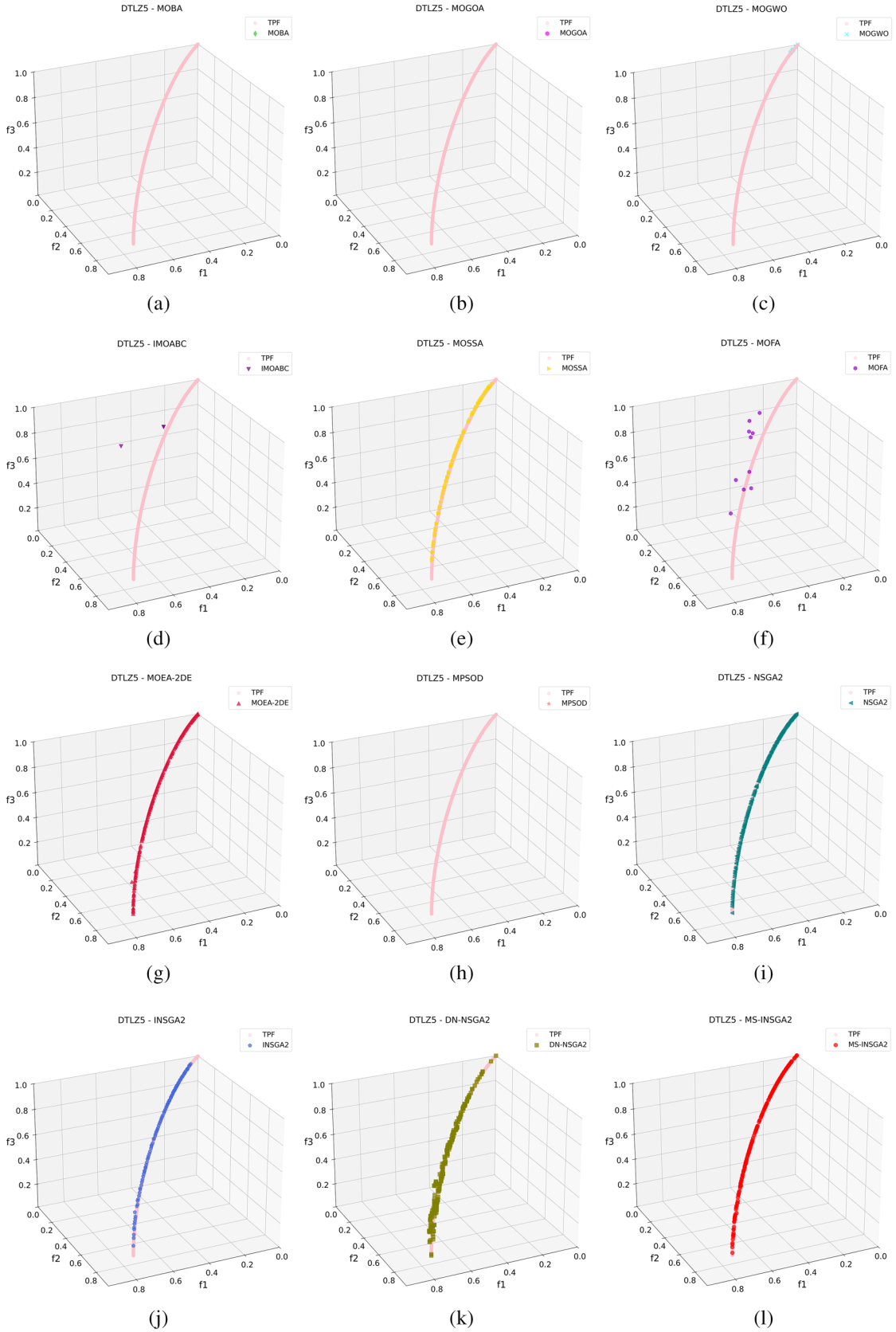


Fig. 8 DTLZ5 represent (*a-l*)

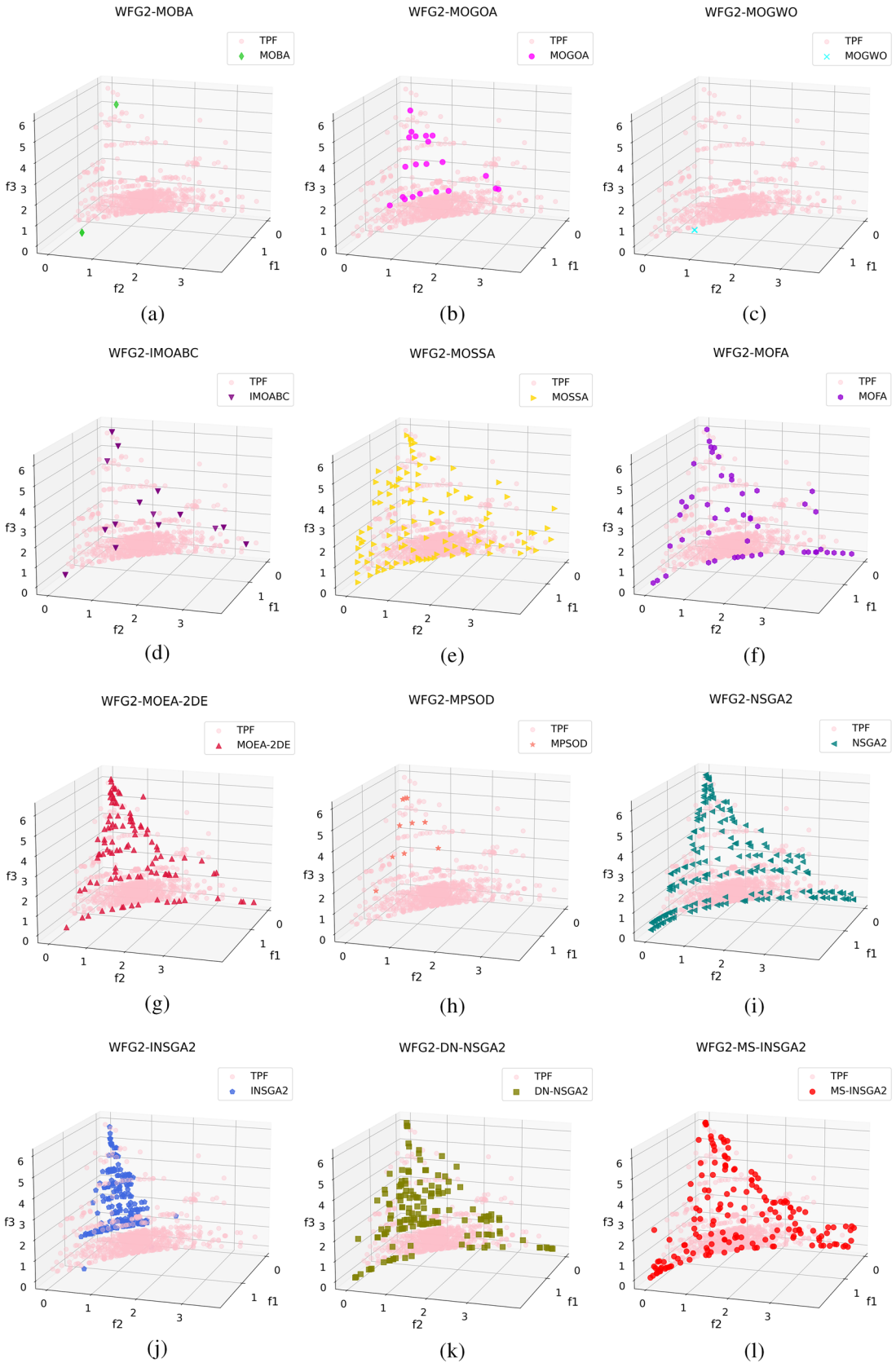


Fig. 9 WFG2 represent (*a-l*)

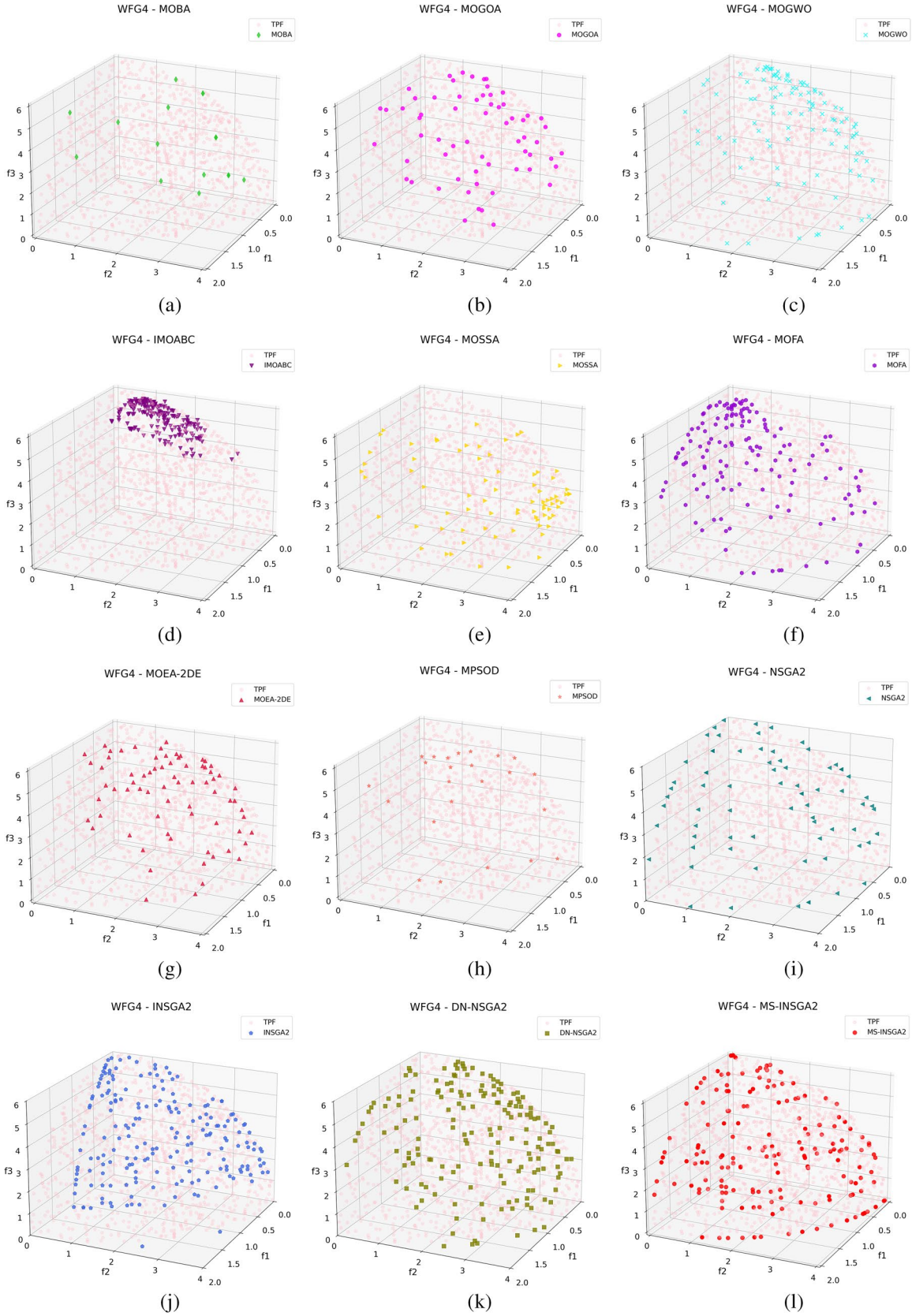


Fig. 10 WFG4 represent (*a-l*)

Table 6 Performance metrics comparison on the ZDT benchmark functions (1)

Problem	Indicator	MOBA		MOGOA		MOGWO		IMOABC		MOSSA		MOFA		MS-INSGA2
ZDT1	IGD	2.0742E−01	−	2.1993E−01	−	2.2775E−01	−	1.8518E−01	−	1.0670E−03	−	9.5053E−02	−	**4.6200E−04**
	HD	2.0742E+00	−	3.2019E+00	−	2.7034E+00	−	3.6434E+00	−	1.3051E−02	−	1.3355E+00	−	**8.4020E−03**
	Spacing	3.6419E+00	−	9.5017E−02	−	1.4648E−01	−	5.6466E−02	−	3.6290E−03	−	1.6677E−01	−	**2.8960E−03**
ZDT2	IGD	3.4810E−01	−	3.2779E−01	−	3.7039E−01	−	7.3351E−02	−	1.0750E−02	−	7.4137E−02	−	**3.0700E−04**
	HD	4.0101E+00	−	3.7954E+00	−	3.8592E+00	−	7.3352E−01	−	2.7651E−02	−	7.4140E−01	−	**4.8690E−03**
	Spacing	1.4553E+00	−	4.3474E−01	−	1.5870E−01	−	5.4985E−02	−	8.7980E−03	−	1.0867E−02	−	**4.1160E−03**
ZDT3	IGD	2.2270E−02	−	3.4061E−02	−	2.4617E−02	−	5.6560E−03	−	2.0700E−04	−	9.4001E−03	−	**1.0200E−04**
	HD	1.1481E+00	−	2.4237E+00	−	1.2692E+00	−	2.9158E−01	−	1.0677E−02	−	4.8464E−01	−	**5.2610E−03**
	Spacing	3.2512E−01	−	7.6905E−02	−	**3.6700E−04**	+	4.6946E−02	−	6.3880E−03	−	8.2570E−03	−	4.7880E−03
ZDT4	IGD	2.8361E+00	−	4.7341E+01	−	1.0142E+00	−	4.6676E+00	−	8.5730E−03	−	7.6074E+01	−	**7.9430E−03**
	HD	3.8972E+00	−	1.2346E+02	−	1.0143E+00	−	3.0089E+01	−	**8.5732E−03**	+	1.1393E+02	−	3.9361E−02
	Spacing	7.7283E−01	−	1.2242E+01	−	**3.7850E−03**	+	2.8901E+00	−	4.5190E−03	+	2.8106E+01	−	4.9560E−03
ZDT6	IGD	2.3319E−02	−	1.7425E−01	−	8.1270E−03	−	6.1600E−04	−	**1.2900E−04**	+	6.4780E−02	−	1.3500E−04
	HD	7.3740E−01	−	6.7477E+00	−	2.5700E−01	−	1.9476E−02	−	4.0890E−03	−	2.3422E+00	−	**2.5170E−03**
	Spacing	3.2628E−01	−	1.8047E−01	−	7.2920E−02	−	1.1233E−02	−	3.0630E−03	−	1.7427E−01	−	**2.5120E−03**
W +/−/≈	IGD	0/5/0		0/5/0		0/5/0		0/5/0		1/4/0		0/5/0		
	HD	0/5/0		0/5/0		0/5/0		0/5/0		1/4/0		0/5/0		
	Spacing	0/5/0		0/5/0		2/3/0		0/5/0		1/4/0		0/5/0		

Table 7 Performance metrics comparison on the ZDT benchmark functions (2)

Problem	Indicator	MOEA-2DE		MPSOD		NSGA2		INSGA2		DN-NSGA2		MS-INSGA2
ZDT1	IGD	1.1479E−01	−	1.2879E−01	−	8.9800E−04	−	2.9660E−03	−	6.9000E−04	−	**4.6200E−04**
	HD	1.2716E+00	−	1.2879E+00	−	9.0250E−03	−	2.9655E−02	−	**8.1630E−03**	+	8.4020E−03
	Spacing	4.6675E−02	−	2.3231E−02	−	6.4470E−03	−	8.6700E−03	−	3.7420E−03	−	**2.8960E−03**
ZDT2	IGD	1.7721E−01	−	7.3352E−02	−	7.3468E−02	−	2.9610E−03	−	8.3900E−04	−	**3.0700E**−04
	HD	1.8289E+00	−	7.3352E−01	−	7.3469E−01	−	2.9608E−02	−	8.3910E−03	−	**4.8690E**−03
	Spacing	9.2197E−02	−	1.7590E−02	−	**2.7090E−03**	+	9.4190E−03	−	6.3820E−03	−	4.1160E−03
ZDT3	IGD	1.7714E−02	−	1.9300E−02	−	4.7410E−03	−	1.3800E−04	−	1.3002E−04	−	**1.0200E−04**
	HD	9.1325E−01	−	9.9504E−01	−	2.4442E−01	−	7.3480E−03	−	6.7160E−03	−	**5.2610E−03**
	Spacing	5.3319E−02	−	**1.3460E−03**	+	5.0080E−03	−	6.9340E−03	−	5.0630E−03	−	4.7880E−03
ZDT4	IGD	1.9304E+01	−	7.4857E+02	−	8.1680E−03	−	1.8541E+00	−	**5.3110E−03**	+	7.9430E−03
	HD	3.5109E+01	−	8.9293E+02	−	**1.5290E−02**	+	3.8586E+00	−	8.8395E−02	−	3.9361E−02
	Spacing	2.3163E+00	−	3.3945E+01	−	6.7140E−03	−	3.2502E−01	−	**4.1780E−03**	+	4.9560E−03
ZDT6	IGD	1.4300E−03	−	7.1577E−03	−	2.4040E−03	−	1.6900E−04	−	1.0630E−03	−	**1.3500E−04**
	HD	4.5218E−02	−	3.3553E+00	−	1.2896E−01	−	5.2527E−02	−	3.3627E−02	−	**2.5170E−03**
	Spacing	1.0253E−02	−	1.4782E−01	−	6.4423E−02	−	3.6558E−02	−	5.8180E−03	−	**2.5120E−03**
W+/−/≈	IGD	0/5/0		0/5/0		0/5/0		0/5/0		1/4/0		
	HD	0/5/0		0/5/0		1/4/0		0/5/0		1/4/0		
	Spacing	0/5/0		1/4/0		1/4/0		0/5/0		1/4/0		

characteristics, deep strategic coupling, and algorithmic limitations:

Problem-oriented analysis based on benchmark characteristics: MS-INSGA2 exhibits targeted optimization advantages across different test suites. When tackling the ZDT test suite, which is characterized by local optima traps, the algorithm's superiority primarily stems from the simulated annealing (SA)-based local search mechanism applied to the 20% elite individuals in the Pareto front. Its Metropolis criterion effectively assists the population in escaping local extrema with a certain probability. When processing the DTLZ test suite featuring high-dimensional objective spaces, the diversity-feedback-based adaptive mutation (Eq. 8) acts as an "emergency response" mechanism; when facing severe diversity depletion caused by dimensional crowding, it rapidly increases the mutation rate to maintain the expansiveness of the population distribution. Conversely, when dealing with the WFG test suite characterized by highly deceptive landscapes, the dynamic guidance strategy utilizing the Pareto front mean (Eq. 5) provides robust, correct directional traction during the early iterations (when the guidance strength s is high), effectively resisting the interference of deceptive attractors on the population's evolutionary direction.

Closed-loop coupling and interaction of innovative strategies: The superiority of the algorithm is not a simple superposition of individual strategies, but originates from deep underlying synergy. On the one hand, the “safety-verified linear interpolation initialization” provides the population with a preliminary “feasible solution baseline” that traverses threat zones. This fundamentally shifts the computational burden of the subsequent "adaptive crossover and mutation" from “blindly searching for valid paths in the entire space” to “meticulously enhancing path quality.” On the other hand, the dynamic guidance strategy and adaptive mutation constitute a rigorous dynamic feedback closed-loop: the former acts as an "accelerator" for convergence, while the diversity metric D serves as a “sensor”. Once excessive crowding is detected ($D < 0.15$), it immediately triggers a random perturbation and mutation mechanism following a uniform distribution $U(-200,200)$, functioning as a “brake” to prevent premature convergence.

Algorithmic trade-offs and potential limitations: Although the proposed algorithm exhibits excellent comprehensive performance, objective trade-offs and limitations still exist in practical applications. Regarding the computational overhead trade-off, the introduction of a maximum 20-step SA local search per iteration for the 20% elite individuals comes at the expense of increased fitness evaluation time, exchanged for the high precision of the Pareto front distribution (i.e., the trade-off between solution quality and time cost). In terms of topological environmental adaptability limitations, when facing extreme scenarios where there is absolutely no line-of-sight path and obstacles are densely distributed like a "maze," the linear interpolation initialization mechanism, which relies solely on large-step perturbations, may struggle to rapidly construct a completely collision-free initial trajectory in the first generation, thereby temporarily restricting the early-stage search efficiency of the algorithm.

Table 8 Performance metrics comparison on the DTLZ benchmark functions (1)

Problem	Indicator	MOBA		MOGOA		MOGWO		IMOABC		MOSSA		MOFA		MS-INSGA2
DTLZ1	IGD	3.2294E+01	−	3.6889E+01	−	4.9493E+01	−	3.0360E−02	−	6.2248E+00	−	2.5545E+01	−	**2.3122E−02**
	HD	4.6539E+01	−	5.0477E+01	−	6.2181E+01	−	8.1997E−02	−	5.3079E+01	−	2.3366E+02	−	**8.0116E−02**
	Spacing	4.0092E+00	−	3.7198E+00	−	6.5701E−01	−	**1.3755E−02**	+	2.9240E+00	−	3.9714E+01	−	1.4525E−02
DTLZ2	IGD	2.2313E−02	−	2.8299E−02	−	2.8468E−02	−	7.0730E−03	−	3.8690E−03	−	7.6790E−03	−	**1.9880E−03**
	HD	7.8837E−01	−	1.4493E+00	−	9.0023E−01	−	2.2366E−01	−	3.2754E−01	−	2.4284E−01	−	**6.2857E−02**
	Spacing	1.9738E−01	−	6.0598E−01	−	**8.8780E−03**	+	9.5726E−02	−	1.6309E−01	−	7.8951E−02	−	2.3979E−02
DTLZ3	IGD	2.0358E+02	−	1.7914E+02	−	2.4366E+02	−	4.9285E+00	−	1.7221E+01	−	3.9939E+02	−	**5.7013E−01**
	HD	2.2671E+02	−	2.6382E+02	−	3.3445E+02	−	1.3472E+01	−	6.0614E+01	−	7.8754E+02	−	**1.1078E−01**
	Spacing	2.9171E+01	−	4.7562E+01	−	2.5812E+01	−	1.2424E+00	−	3.3402E+01	−	1.8243E+02	−	**3.0700E−02**
DTLZ4	IGD	4.4585E−02	−	7.5399E−02	−	3.4508E−02	−	4.4191E−02	−	**3.1830E−03**	+	2.0147E−02	+	2.4585E−02
	HD	1.4099E+00	−	2.3843E+00	−	1.0913E+00	−	1.3974E+00	−	**2.6423E−01**	+	6.3710E−01	−	5.3129E−01
	Spacing	5.2800E−01	−	1.0672E−01	−	1.0911E+00	−	1.1394E−02	−	1.1422E−01	−	5.2997E−01	−	**6.4590E−03**
DTLZ5	IGD	1.8469E−02	−	1.8175E−02	−	2.3627E−02	−	1.9711E−02	−	7.2500E−03	−	1.0244E−02	−	**2.3600E−04**
	HD	5.8405E−01	−	5.7475E−01	−	7.4717E−01	−	6.2330E−01	−	2.2915E−02	−	3.2396E−01	−	**5.4500E−03**
	Spacing	7.8587E−02	−	9.2127E−01	−	8.2840E−03	−	2.3834E−02	−	6.9060E−03	−	2.8874E−02	−	**2.3660E−03**
DTLZ6	IGD	4.9700E−04	−	7.2047E−02	−	2.6964E−02	−	6.3332E−02	−	2.1900E−04	−	1.0665E−01	−	**2.0700E−04**
	HD	1.5719E−02	−	4.2899E+00	−	8.5266E−01	−	4.3335E+00	−	6.9150E−03	−	4.4423E+00	−	**3.3790E−03**
	Spacing	9.3020E−03	−	6.2905E−01	−	9.3947E−02	−	3.5835E−01	−	5.3900E−03	−	7.0246E−01	−	**2.6260E−03**
DTLZ7	IGD	5.1489E−02	−	2.8786E−01	−	4.9597E−02	−	4.6626E−02	−	1.4347E−02	−	3.6586E−02	−	**1.7300E−03**
	HD	1.6477E+00	−	9.2116E+00	−	1.5870E+00	−	1.4920E+00	−	4.5909E+00	−	1.1708E+00	−	**5.5374E−02**
	Spacing	1.8698E+00	−	2.1857E−01	−	1.7800E+00	−	**1.8410E−03**	+	5.6962E−02	−	1.8875E−02	+	2.2876E−02
W+/−/≈	IGD	0/7/0		0/7/0		0/7/0		0/7/0		1/6/0		1/6/0		
	HD	0/7/0		0/7/0		0/7/0		0/7/0		1/6/0		0/7/0		
	Spacing	0/7/0		0/7/0		1/6/0		2/5/0		0/7/0		1/6/0		

Table 9 Performance metrics comparison on the DTLZ benchmark functions (2)

Problem	Indicator	MOEA-2DE		MPSOD		NSGA2		INSGA2		DN-NSGA2		MS-INSGA2
DTLZ1	IGD	2.9297E−02	−	1.8168E+01	−	2.9193E−01	−	1.2968E+01	−	4.0277E−02	−	**2.3122E−02**
	HD	8.2002E−02	−	5.4545E+01	−	5.6842E−01	−	1.6754E+02	−	4.2962E+00	−	**8.0116E−02**
	Spacing	1.1233E−02	+	1.3127E+00	−	**1.1184E−02**	+	2.4808E+01	−	1.7239E−02	−	1.4525E−02
DTLZ2	IGD	1.7090E−03	+	1.4662E−02	−	**1.6290E−03**	+	7.9340E−03	−	2.8810E−03	−	1.9880E−03
	HD	5.4030E−02	+	5.2364E−01	−	**5.1523E−02**	+	2.5089E−01	−	9.1112E−02	−	6.2857E−02
	Spacing	3.3106E−02	−	2.0054E−01	−	2.7556E−02	−	1.1379E−01	−	3.4412E−02	−	**2.3979E−02**
DTLZ3	IGD	6.1371E+04	−	**2.2502E−02**	+	1.2198E+01	−	1.0821E+02	−	1.6796E−01	+	5.7013E−01
	HD	6.5298E+00	−	9.3260E−01	−	5.7565E+01	−	3.2642E+02	−	1.7268E−01	−	**1.1078E−01**
	Spacing	2.3722E−01	−	2.5389E−01	−	7.8909E+00	−	1.0327E+02	−	**2.8972E−02**	+	3.0700E−02
DTLZ4	IGD	3.4502E−02	−	3.4485E−02	−	3.3125E−02	−	**1.4885E−02**	+	3.4502E−02	−	2.4585E−02
	HD	1.0910E+00	−	1.3761E+00	−	1.0475E+00	−	**4.7071E−01**	+	8.5225E−01	−	5.3129E−01
	Spacing	8.7201E−02	−	2.0553E−01	−	2.8619E−02	−	2.7681E−01	−	3.1549E−02	−	**6.4590E−03**
DTLZ5	IGD	6.0900E−04	−	1.3463E−02	−	**2.0900E−04**	+	6.8300E−04	−	4.3800E−04	−	2.3600E−04
	HD	1.9263E−02	−	4.2574E−01	−	6.5990E−03	−	2.1583E−02	−	1.5276E−02	−	**5.4500E−03**
	Spacing	6.4800E−03	−	1.7876E−02	−	4.1420E−03	−	2.4940E−03	−	9.9980E−03	−	**2.3660E−03**
DTLZ6	IGD	7.5004E−04	−	1.9140E−03	−	**1.2100E−04**	+	2.5620E−01	−	1.2900E−04	+	2.0700E−04
	HD	3.6900E−03	−	6.0531E−02	−	3.8130E−03	−	8.4499E+00	−	4.0860E−03	−	**3.3790E−03**
	Spacing	3.6070E−03	−	2.6350E−02	−	3.9110E−03	−	2.8668E−01	−	3.6840E−03	−	**2.6260E−03**
DTLZ7	IGD	6.6403E−02	−	5.5183E−02	−	2.1210E−03	−	3.6118E−02	−	9.1370E−03	−	**1.7300E−03**
	HD	2.1249E+00	−	1.7659E+00	−	6.7868E−02	−	1.1558E+00	−	2.9240E−01	−	**5.5374E−02**
	Spacing	1.1881E−01	−	5.6935E−02	−	3.2731E−02	−	8.1936E−02	−	3.5775E−02	−	**2.2876E−02**
W +/−/≈	IGD	1/6/0		1/6/0		3/4/0		1/6/0		2/5/0		
	HD	1/6/0		0/7/0		1/6/0		1/6/0		0/7/0		
	Spacing	1/6/0		0/7/0		1/6/0		0/7/0		1/6/0		

5 UAV path planning experiment

5.1 Task scheduling

To comprehensively verify the effectiveness and stability of the proposed multi-objective MS-INSGA2 algorithm, a 3D complex terrain simulation environment with dimensions of 3500 m × 3500 m × 120 m is constructed in this study. The algorithm simulation experiments are executed on a computing platform equipped with an Intel(R) Core(TM) i9-14900K (3.20 GHz) processor and 48GB of RAM. All algorithm code development and testing are conducted within the PyCharm 2025.2.4 integrated development environment (IDE). Based on this environment, two sets of systematic comparative experiments are designed, with the detailed parameters presented in Table 12.

Specifically, Task I is an ablation study based on the control variable method, aiming to quantitatively evaluate the independent contribution of each improvement mechanism. This experiment conducts an in-depth comparison among the complete MS-INSGA2 algorithm, the baseline NSGA2 algorithm, and four variant algorithms that each introduce only a single improvement mechanism. The specific definitions of these variant algorithms are as follows: G-INSGA2 incorporates only the dynamic guidance strategy into the NSGA2 algorithm; A-INSGA2 incorporates only the adaptive crossover and mutation strategy; I-INSGA2 incorporates only the interpolation initialization strategy; and L-INSGA2 incorporates only the local search strategy. Task II serves as a comprehensive performance comparative experiment, which comprises two sub-tasks: Comprehensive I and Comprehensive II. All tasks are independently iterated 100 times, and the entire experiment is repeated 20 times to ensure the statistical significance and robustness of the experimental results.

5.2 Multi-objective function

5.2.1 Path length

The total length of the path planning is set as objective one of the UAV path planning, as shown in Eq. 20. Firstly, the distance of the flight segment is obtained by calculating the Euclidean distance between the two track points, and the total flight track length (Zhang et al. 2025) is solved by combining the cumulative distance of all the flight segments.

Table 10 Performance metrics comparison on the WFG benchmark functions (1)

Problem	Indicator	MOBA		MOGOA		MOGWO		IMOABC		MOSSA		MOFA		MS-INSGA2
WFG1	IGD	6.9240E−02	−	7.5067E−02	−	1.0663E−01	−	8.5108E−02	−	**5.0220E−02**	+	6.8678E−02	−	5.1219E−02
	HD	2.1896E+00	−	2.3738E+00	−	3.3721E+00	−	2.6914E+00	−	1.5881E+00	−	2.1718E+00	−	**1.4963E+00**
	Spacing	7.0672E−02	−	6.4230E−02	−	1.0865E+00	−	1.0501E+00	−	1.1990E−01	−	6.9749E−01	−	**1.7468E−02**
WFG2	IGD	8.2267E−02	−	3.9894E−02	−	5.7849E−02	−	1.1317E−01	−	8.9420E−03	−	1.5902E−02	−	**7.2100E−03**
	HD	2.3181E+00	−	1.1241E+00	−	1.6301E+00	−	3.1890E+00	−	3.4242E−01	−	4.4808E−01	−	**2.1477E−01**
	Spacing	9.8085E−01	−	1.3183E−01	−	**5.4002E−05**	+	5.0074E−02	−	1.3283E−01	−	2.7537E−01	−	1.0600E−02
WFG3	IGD	3.0540E−02	−	2.0414E−02	−	7.7270E−03	−	1.8141E−02	−	3.3870E−03	−	4.4260E−03	−	**2.1480E−03**
	HD	1.1087E+00	−	9.2753E−01	−	**4.9088E−01**	+	8.6122E−01	−	7.0346E−01	−	8.0996E−01	−	6.0841E−01
	Spacing	3.8129E−01	−	9.9965E−02	−	9.2634E−02	−	9.1580E−02	−	7.2367E−02	−	8.3489E−02	−	**4.0783E−02**
WFG4	IGD	2.7648E−02	−	2.0317E−02	−	1.6451E−02	−	6.0748E−02	−	1.6881E−02	−	1.1851E−02	−	**8.4510E−03**
	HD	8.7429E−01	−	6.4247E−01	−	5.2021E−01	−	1.9210E+00	−	5.3382E−01	−	3.7477E−01	−	**2.6724E−01**
	Spacing	4.1142E−01	−	1.4717E−01	−	1.6643E−01	−	**4.1473E−02**	+	1.7984E−01	−	1.2286E−01	−	1.1699E−01
WFG5	IGD	2.7419E−02	−	2.8047E−02	−	1.3414E−01	−	1.2776E−02	−	9.0950E−03	−	1.1843E−02	−	**8.7530E−03**
	HD	8.6707E−01	−	8.8692E−01	−	4.2418E+00	−	4.0402E-01	−	2.8760E−01	−	3.7450E−01	−	**2.7678E−01**
	Spacing	2.2655E−01	−	1.6120E−01	−	**2.8867E−02**	+	1.1619E−01	−	1.2735E−01	−	1.0974E−01	−	9.7765E−02
WFG6	IGD	1.3080E−01	−	2.8063E−02	−	2.1273E−02	−	1.6938E−02	−	**1.0909E−02**	+	1.5527E−02	−	1.2805E−02
	HD	4.1362E+00	−	9.1514E−01	−	6.7272E−01	−	5.3562E−01	−	**3.4498E−01**	+	4.9100E−01	−	4.0494E−01
	Spacing	5.0192E−01	−	1.5767E−01	−	1.4939E−01	−	2.8775E−01	−	1.3237E−01	−	1.4977E−01	−	**1.0020E−01**
WFG7	IGD	9.9123E−02	−	2.1909E−02	−	1.3558E−01	−	1.7276E−02	−	1.2918E−02	−	1.2554E−02	−	**7.2980E−03**
	HD	3.1345E+00	−	6.9281E−01	−	4.2875E+00	−	5.4633E−01	−	4.0849E−01	−	3.9698E−01	−	**2.3078E−01**
	Spacing	3.1776E−01	−	1.8471E−01	−	3.8180E−02	−	2.8204E−01	−	1.4349E−01	−	1.1157E−01	−	**1.4705E−02**
WFG8	IGD	1.5902E−01	−	9.5663E−02	−	4.4380E−02	−	**1.9531E−02**	+	3.3545E−02	−	3.2494E−02	−	2.6534E−02
	HD	1.5973E+00	−	1.0179E+00	−	6.0079E−01	−	3.6499E+00	−	4.7707E−01	−	4.4674E−01	−	**3.2863E−01**
	Spacing	8.2202E−01	−	1.3900E−01	−	1.5527E−01	−	2.7483E−01	−	1.4784E−01	−	1.6739E−01	−	**1.0804E−01**
WFG9	IGD	1.2603E−01	−	2.5937E−02	−	4.6102E−02	−	2.0742E−02	−	9.6390E−03	−	1.3084E−02	−	**7.4670E−03**
	HD	3.9854E+00	−	8.2021E-01	−	1.4579E+00	−	6.5591E−01	−	3.0480E−01	−	4.1376E−01	−	**2.3613E−01**
	Spacing	**3.8520E−02**	+	2.3702E−01	−	1.2953E−01	−	2.2992E−01	−	1.2150E−01	−	1.5033E−01	−	1.0126E−01
W+/−/≈	IGD	0/9/0		0/9/0		0/9/0		1/8/0		2/7/0		0/9/0		
	HD	0/9/0		0/9/0		1/8/0		0/9/0		1/8/0		0/9/0		
	Spacing	1/8/0		0/9/0		2/7/0		1/8/0		0/9/0		0/9/0		

Table 11 Performance metrics comparison on the WFG benchmark functions (2)

Problem	Indicator	MOEA-2DE		MPSOD		NSGA2		INSGA2		DN-NSGA2		MS-INSGA2
WFG1	IGD	6.3215E−02	−	7.4725E−02	−	5.4683E−02	−	6.4116E−02	−	6.1858E−02	−	**5.1219E−02**
	HD	1.9990E+00	−	2.3630E+00	−	1.7292E+00	−	2.0275E+00	−	1.9561E+00	−	**1.4963E+00**
	Spacing	3.4680E−01	−	3.9474E−02	−	2.0498E−02	−	7.5478E−02	−	3.4349E−02	−	**1.7468E−02**
WFG2	IGD	1.7675E−02	−	3.3774E−02	−	7.5110E−03	−	2.7756E−02	−	8.8230E−03	−	**7.2100E−03**
	HD	4.9806E−01	−	9.5168E−01	−	2.2444E−01	−	7.8210E−01	−	3.1105E−01	−	**2.1477E−01**
	Spacing	1.3095E−01	−	2.1581E−01	−	9.2863E−02	−	3.9521E−02	−	1.2150E−01	−	**1.0600E−02**
WFG3	IGD	1.5991E−02	−	8.4643E−02	−	**1.8860E−03**	+	3.6320E−03	−	5.7830E−03	−	2.1480E−03
	HD	8.4668E−01	−	2.6766E+00	−	9.0439E−01	−	8.8638E−01	−	9.3006E−01	−	**6.0841E−01**
	Spacing	2.4981E−01	−	**1.9234E−02**	+	5.7519E−02	−	5.6919E−02	−	5.0876E−02	−	4.0783E−02
WFG4	IGD	2.0768E−02	−	2.5003E−02	−	1.4114E−01	−	1.1415E−02	−	1.3875E−02	−	**8.4510E−03**
	HD	6.5673E−01	−	7.9066E−01	−	4.4632E+00	−	3.6097E−01	−	4.3875E−01	−	**2.6724E−01**
	Spacing	1.9316E−01	−	1.8798E−01	−	1.6221E−01	−	**1.1431E−01**	+	1.4538E−01	−	1.1699E−01
WFG5	IGD	9.8190E−03	−	2.5437E−02	−	1.0336E−02	−	1.0522E−02	−	1.3447E−02	−	**8.7530E−03**
	HD	3.1051E−01	−	8.0439E−01	−	3.2685E−01	−	3.3275E−01	−	4.2524E−01	−	**2.7678E−01**
	Spacing	1.0450E−01	−	1.3575E−01	−	1.0117E−01	−	1.4673E−01	−	1.1739E−01	−	**9.7765E−02**
WFG6	IGD	1.6903E−02	−	2.6645E−02	−	1.4667E−02	−	1.6460E−02	−	**1.2180E−02**	+	1.2805E−02
	HD	5.6056E−01	−	8.4259E−01	−	**2.4235E−01**	+	5.2052E−01	−	3.8515E−01	+	4.0494E−01
	Spacing	1.2606E−01	−	1.4914E−01	−	1.1513E−01	−	1.6014E−01	−	1.1515E−01	−	**1.0020E−01**
WFG7	IGD	2.4872E−02	−	6.6404E−02	−	1.1043E−02	−	8.9320E−03	−	1.6274E−02	−	**7.2980E−03**
	HD	7.8651E−01	−	2.0999E+00	−	3.4922E−01	−	2.8245E−01	−	5.1463E−01	−	**2.3078E−01**
	Spacing	4.0580E−01	−	**1.1648E−02**	+	1.1874E−01	−	8.9030E−02	−	1.2411E−01	−	1.4705E−02
WFG8	IGD	3.9517E−02	−	3.6455E−02	−	**2.0443E−02**	+	2.3197E−02	+	2.9401E−02	−	2.6534E−02
	HD	5.5886E−01	−	8.0178E−01	−	3.7775E−01	−	6.4862E−01	−	5.4082E−01	−	**3.2863E−01**
	Spacing	1.2204E−01	−	1.7188E−01	−	1.2611E−01	−	**9.0128E−02**	+	1.1984E−01	−	1.0804E−01
WFG9	IGD	8.2620E−03	−	8.7013E−02	−	1.1030E−02	−	9.4520E−03	−	1.1973E−02	−	**7.4670E−03**
	HD	2.6126E−01	−	2.7516E+00	−	3.4879E−01	−	2.9889E−01	−	3.7861E−01	−	**2.3613E−01**
	Spacing	1.0095E−01	+	**7.3580E−02**	+	1.1241E−01	−	1.0190E−01	−	1.6273E−01	−	1.0126E−01
W+/−/≈	IGD	0/9/0		0/9/0		2/7/0		1/8/0		1/8/0		
	HD	0/9/0		0/9/0		1/8/0		0/9/0		1/8/0		
	Spacing	1/8/0		3/6/0		0/9/0		2/7/0		0/9/0		

Table 12 Experiment scene arrangement

scene	Experiment	Start coordinates	Target coordinates	Threat points	Route points
Ⅰ	MS-INSGA2 vs (NSGA2、G-INSGA2 、A-INSGA2、I-INSGA2、L-INSGA2)	(200,3200,50)	(3200,500,50)	5	5
		(200,3200,50)	(3200,500,50)	5	5
		(200,3200,50)	(3200,500,50)	5	5
		(200,3200,50)	(3200,500,50)	5	5
		(200,3200,50)	(3200,500,50)	5	5
Ⅱ	Comprehensive Ⅰ	(200,3200,50)	(3200,500,50)	5	5
	Comprehensive Ⅱ	(200,3200,60)	(3200,500,60)	10	5

$$F_1 = \sum_{i=1}^{N-1} \sqrt{\left(x_{i+1} - x_i\right)^2 + \left(y_{i+1} - y_i\right)^2 + \left(z_{i+1} - z_i\right)^2} \quad (20)$$

In Formula 20, x_i、y_i and z_i respectively represent the positions of the i-th waypoint on the x-axis, y-axis, and z-axis.

5.2.2 Collision risk

Collision risk is set as the second objective of UAV path planning, as shown in Formula 21. It mainly consists of two components: terrain risk F_{terrain} (Wang and Tian 2026) and current cylindrical obstacle risk $F_{cylinder}$ (Gu et al. 2025).

$$F_2 = F_{terrain} + F_{cylinder} \tag{21}$$

The terrain collision risk is mainly judged by whether the difference between the track height of the UAV flight and the terrain height is within a reasonable safety threshold, as shown in Eq. 22.

$$F_{terrain} = \sum_{i=1}^{N} \begin{cases} 200 \cdot \frac{h_{terrain}(x_i,y_i)+50-z_i}{50}, h_{terrain}(x_i, y_i) + 50 > z_i \\ 0, others \end{cases} \tag{22}$$

In Formula 22, $h_{terrain}(x_i,y_i)$ denotes the terrain height corresponding to the waypoint (x_i,y_i),z_i represents the height coordinate of the i-th waypoint, and N denotes the total number of waypoints. Corresponding to the terrain collision risk, there is also the cylindrical obstacle risk encountered by the current UAV during flight. The risk is divided into 5 layers based on the distance between the UAV and the cylindrical obstacle, as shown in Formula 23 and Fig. 11.

$$F_{cylinder} = \sum_{r=1}^{R} \sum_{i=1}^{N} \begin{cases} 1500, d_{level}(i,r) \le r_{cylinder} \wedge z_i \in [r_{z,bottom}, r_{z,top}] \\ 300 \cdot \frac{r_{cylinder}+200-d_{horiz}(i,r)}{200}, r_{cylinder} < d_{level}(i,r) \le r_{cylinder} + 200 \wedge d_p > 0.8 \\ 200 \cdot \frac{r_{cylinder}+200-d_{horiz}(i,r)}{200}, r_{cylinder} < d_{level}(i,r) \le r_{cylinder} + 200 \wedge 0.5 < d_p \le 0.8 \\ 100 \cdot \frac{r_{cylinder}+200-d_{horiz}(i,r)}{200}, r_{cylinder} < d_{level}(i,r) \le r_{cylinder} + 200 \wedge d_p \le 0.5 \\ 0, others \end{cases} \tag{23}$$

In Formula 22, R denotes the total number of cylindrical obstacles, and N represents the total number of waypoints; $\sum_{r=1}^{R} \sum_{i=1}^{N}$ means calculating and summing the risks for each cylindrical obstacle and each waypoint one by one. $r_{cylinder}$ represents the radius of the r-th cylindrical obstacle, and z_i denotes the height coordinate of the i-th waypoint. $r_{z,bottom}$ stands for the bottom height of the r-th cylindrical obstacle, which is $h_{terrain}(r_x,r_y)+1$; r_z,top represents the top height of the r-th cylindrical obstacle, which is $r_{z,bottom+120}$=. d_{horiz} (i,r) denotes the horizontal distance from the i-th waypoint to the r-th cylindrical obstacle, with the model: $\sqrt{(x_i - r_x)^2 + (y_i - r_y)^2}$.$d_p$ represents the proximity of the waypoint to the cylindrical obstacle, and its model is:$\frac{r_{cylinder}+200-d_{horiz}(i,r)}{200}$.

5.2.3 Track smoothness

Track smoothness (Hu et al. 2026) is set as the third goal of UAV track planning, and the smoothness is quantified by using the cumulative sum of the absolute difference of curvature of adjacent segments, as shown in Eq. 24.

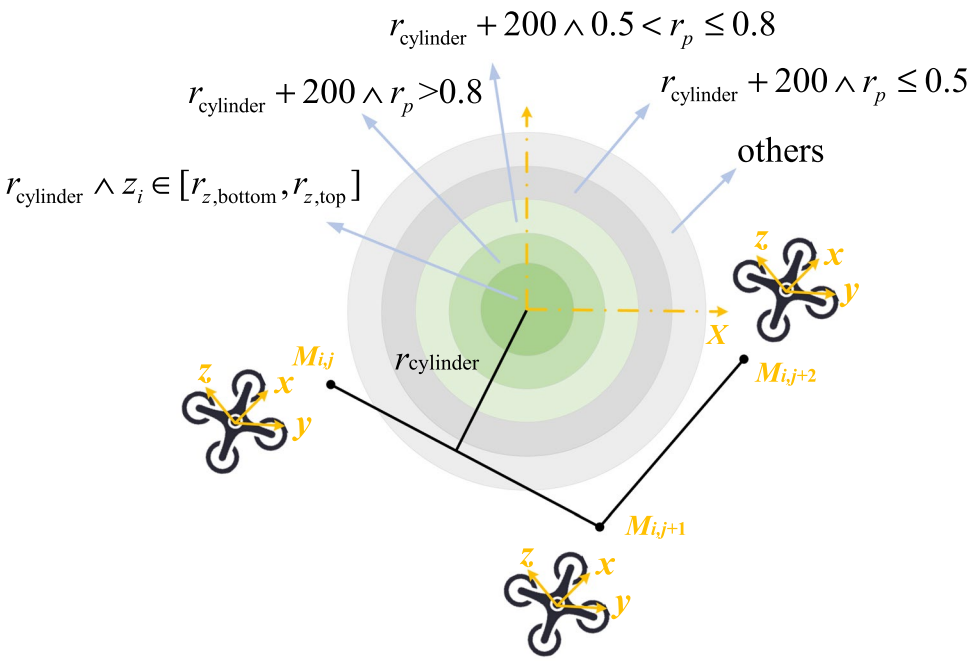


Fig. 11 Cylindrical obstacle risk diagram

$$F_3 = \sum_{i=1}^{N-3} \left| k_{i+1} - k_i \right| \tag{24}$$

In Formula 24, k_i and k_{i+1} denote the curvature of the i-th and $(i+1)$-th trajectory segments, respectively, and the model of curvature k_i is shown in Formula 25.

$$k_i = \frac{\|\vec{v}_i \times \vec{a}_i\|}{\|\vec{v}_i\|^3} \tag{25}$$

In Formula 25, $\vec{v}_i$ refers to the first-order velocity difference of the i-th trajectory segment, defined as:$\vec{v}_i = (x_{i+1} - x_i, y_{i+1} - y_i, z_{i+1} - z_i)$. For the i-th trajectory segment, $\vec{a}_i$ denotes its second-order acceleration difference, given by: $\vec{a}_i = \vec{v}_{i+1} - \vec{v}_i$. $\vec{v}_i \times \vec{a}_i$ stands for the cross product of velocity vector $\vec{v}_i$ and acceleration vector $\vec{a}_i$, which characterizes the perpendicularity between $\vec{v}_i$ and $\vec{a}_i$ while reflecting the rate of change in the velocity direction; $\|\vec{v}_i\|$ refers to the magnitude of the velocity vector $\vec{v}_i$.

5.3 Evaluation metrics for UAV path planning

5.3.1 Generational distance

The Generational Distance (GD)(Mirjalili et al. 2017) calculation model is shown in Eq. 26, which is mainly used to evaluate the distance between the Pareto theoretical front and the true front in the nondominated solution set. The smaller obtained value means that the obtained solution is closer to the true Pareto front value and the convergence is better.

$$GD = \frac{\sqrt{\sum_{i=1}^{n_0} d_i^2}}{n_0} \tag{26}$$

In Formula 26, n_0 denotes the number of optimal solutions obtained on the True Pareto Front (TPF), while d_i refers to the distance between the i-th non-dominated sorted solution and its theoretically closest point.

5.3.2 Hypervolume index

The calculation model of Hypervolume index (HV)(Liang et al. 2024) is shown in Formula 27. The larger the volume is, the more comprehensive the algorithm is proved. The volume of the region in the objective space enclosed by the non-dominated solution set and the reference point was obtained.

$$HV = hypervolume\left(\bigcup_{i=1}^{|P|} S_i \right) \tag{27}$$

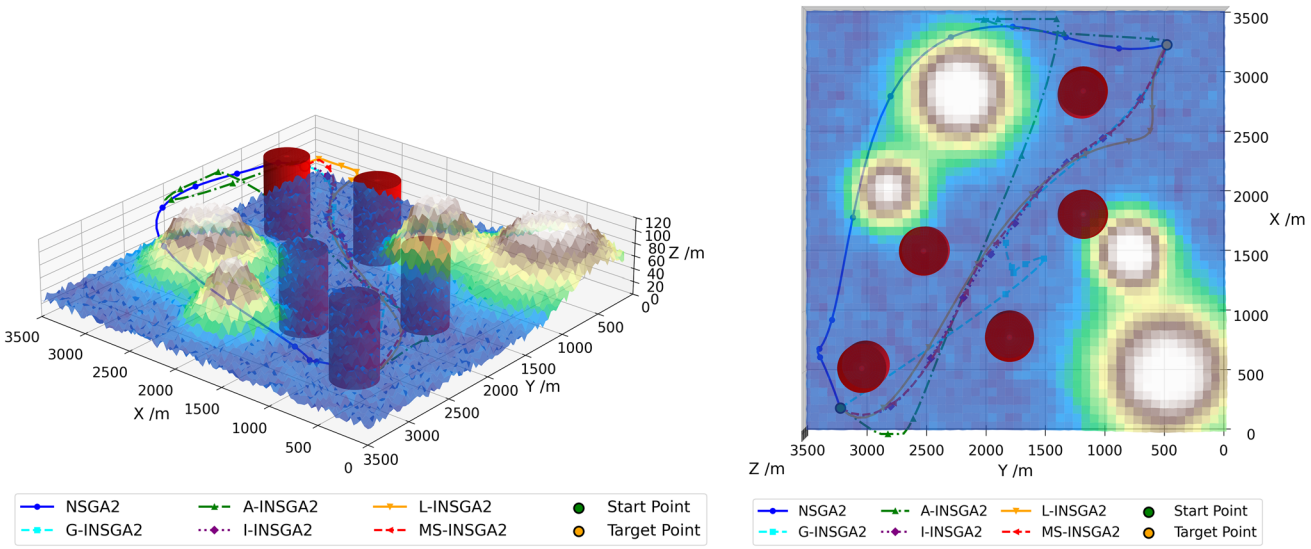
In Formula 27, $|P|$ denotes the number of solutions in the True Pareto Front (TPF) solution set; S_i refers to the set of solutions within the i-th solution group; $\bigcup_{i=1}^{|P|} S$ represents the union of all S_i sets where i ranges from 1 to $|P|$.

5.4 Scene task Ⅰ

Based on the parameter configurations for Scene Task I in Table 12, an ablation study utilizing the control variable method(Deng et al. 2026) was conducted, as depicted in Fig. 12, aiming to quantitatively evaluate the independent contributions of each improvement mechanism to the overall algorithmic performance. This experiment conducts an in-depth comparative analysis among the complete MS-INSGA2 algorithm, the baseline NSGA2 algorithm, and four variant algorithms that each incorporate only a single improvement mechanism. The specific definitions of the variant algorithms are detailed as follows: G-INSGA2 denotes the integration of only the dynamic guidance strategy into the NSGA2 algorithm; A-INSGA2 denotes the integration of only the adaptive crossover and mutation strategy; I-INSGA2 denotes the integration of only the interpolation initialization strategy; and L-INSGA2 denotes the integration of only the local search strategy. In the experimental scenario, the number of intermediate control points for trajectory planning is set to 5. The specific coordinate configuration of the 3D spatial environment is as follows: the green starting point is located at (200, 3200, 50), and the orange target endpoint is located at (3200, 500, 50). Furthermore, five red cylindrical threat obstacles are distributed within the space, with their center coordinates set at (550, 3000, 120), (800, 1800, 120), (1500, 2500, 120), (1800, 1200, 120), and (2800, 1200, 120), respectively.

The results of the ablation study for the MS-INSGA2 algorithm are presented in Fig. 13. As observed from the convergence curves in Figs. 13(a) and 13(b), compared with the baseline NSGA2 algorithm and the four variants incorporating only a single strategy (i.e., G-INSGA2, A-INSGA2, I-INSGA2, and L-INSGA2), the MS-INSGA2 algorithm, which fully integrates all proposed mechanisms, exhibits a significant advantage in the convergence speed of both the average fitness and the best fitness. Furthermore, Fig. 13c indicates that the final fitness values of MS-INSGA2 are situated entirely within the lowest interval, featuring a highly concentrated distribution and an extremely narrow range of dispersion. These results thoroughly verify the strong positive synergistic effects among the multiple improvement strategies proposed in this study, thereby substantiating the effectiveness of the hybrid optimization framework.

As presented in Table 13, the improved hybrid strategy algorithm MS-INSGA2 exhibits the global optimal performance across all evaluation metrics. Compared with the baseline NSGA2 algorithm, the final Best Fitness and Average Fitness of MS-INSGA2 are significantly reduced by



(a) First-person perspective

(b) Top view perspective

Fig. 12 Task Ⅰ: 3D environmental feature distribution

24.57% and 51.36%, respectively. At the physical trajectory level, the planned Path length is shortened by 18.69%, and the Collision risk is substantially mitigated. These results verify the algorithm's outstanding comprehensive performance in terms of optimization precision and trajectory safety.

Furthermore, compared to the four ablation variant algorithms that incorporate only a single strategy—namely, the variant with only the improved initial solution (I-INSGA2), only the local search (L-INSGA2), only the adaptive crossover and mutation (A-INSGA2), and only the dynamic guidance (G-INSGA2)—the Best Fitness of the complete MS-INSGA2 is further reduced by 4.31%, 0.50%, 15.42%, and 35.46%, respectively. Regarding the Average Fitness, its advantage in synergistic optimization is even more prominent, exhibiting reductions of 46.61%, 7.21%, 49.89%, and 46.72%, respectively, compared to the aforementioned variants. These results thoroughly substantiate the significant positive synergistic effects among the four improvement strategies proposed in this study.

5.5 Scene task Ⅱ

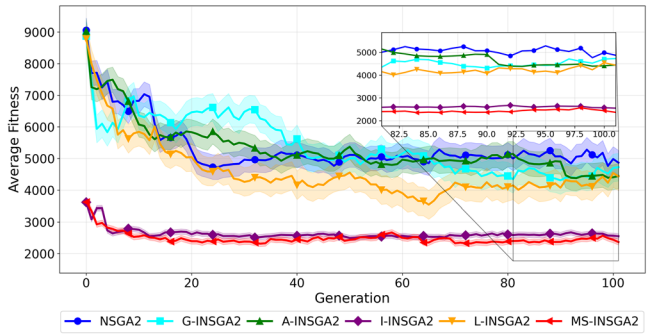
In order to further verify the robustness of the improved hybrid MS-INSGA2 algorithm, MOBA, MOGOA, MOGWO, IMOABC, MOSSA, MOFA, MOEA-2DE, MPSOD, NSGA2, INSGA2 and DN-NSGA2 algorithms were added for comparison. For details, see Comprehensive Ⅰ and Comprehensive Ⅱ experiments.

5.5.1 Comprehensive Ⅰ

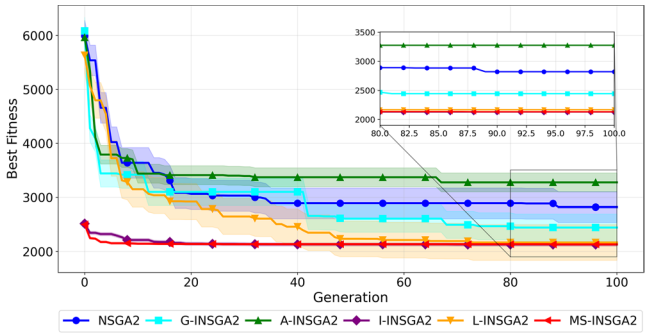
According to the task arrangement of Scene task Ⅱ in Table 12, the Comprehensive Ⅰ experiment is carried out as shown in Fig. 14. The coordinates of the green starting point are set as (200,3200,50), the coordinates of the orange target end point are set as (3200,500,50), and the number of red cylindrical obstacles is set as 5. The middle track points of the path planning are set to 5, and the positions are (550,3000,120), (800,1800,120), (1500,2500,120), (1800,1200,120), (2800,1200,120) respectively.

The results of the Comprehensive Ⅰ experiment are shown in Fig. 15. It is concluded from Fig. 15 (a) and (b) that the improved hybrid MS-INSGA2 algorithm has the best performance in average convergence and optimal convergence speed compared with other algorithms, and it is concluded from Fig. 15 (c) that the hybrid MS-INSGA2 algorithm is robust.

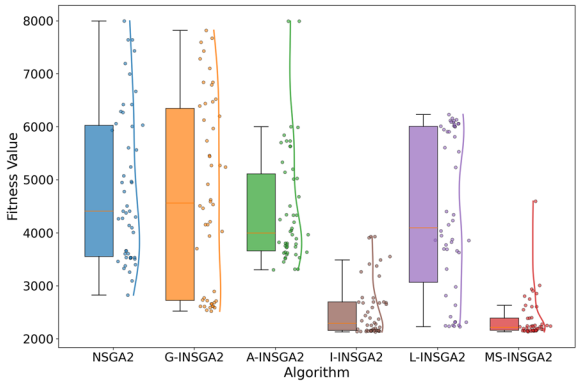
The comparative results between the improved hybrid strategy MS-INSGA2 algorithm and eleven algorithms—namely MOBA, MOGOA, MOGWO, IMOABC, MOSSA, MOFA, MOEA-2DE, MPSOD, NSGA2, INSGA2, and DN-NSGA2—are presented in Table 14. Specifically, regarding the Average Fitness, MS-INSGA2 achieves performance improvements of 26.14%, 38.19%, 40.70%, 19.09%, 13.19%, 28.18%, 13.79%, 6.40%, 57.23%, 13.92%, and 7.78%, respectively, with an overall average improvement of 27.87%. In terms of the Best Fitness, the proposed algorithm exhibits improvements of 4.36%, 4.54%, 30.33%,



(a) Average Fitness Convergence Plot



(b) Best Fitness Convergence Plot



(c) Box and violin picture Distribution

Fig. 13 Task Ⅰ: data visualization chart

Table 13 Quantitative evaluation of the ablation study in scene task I

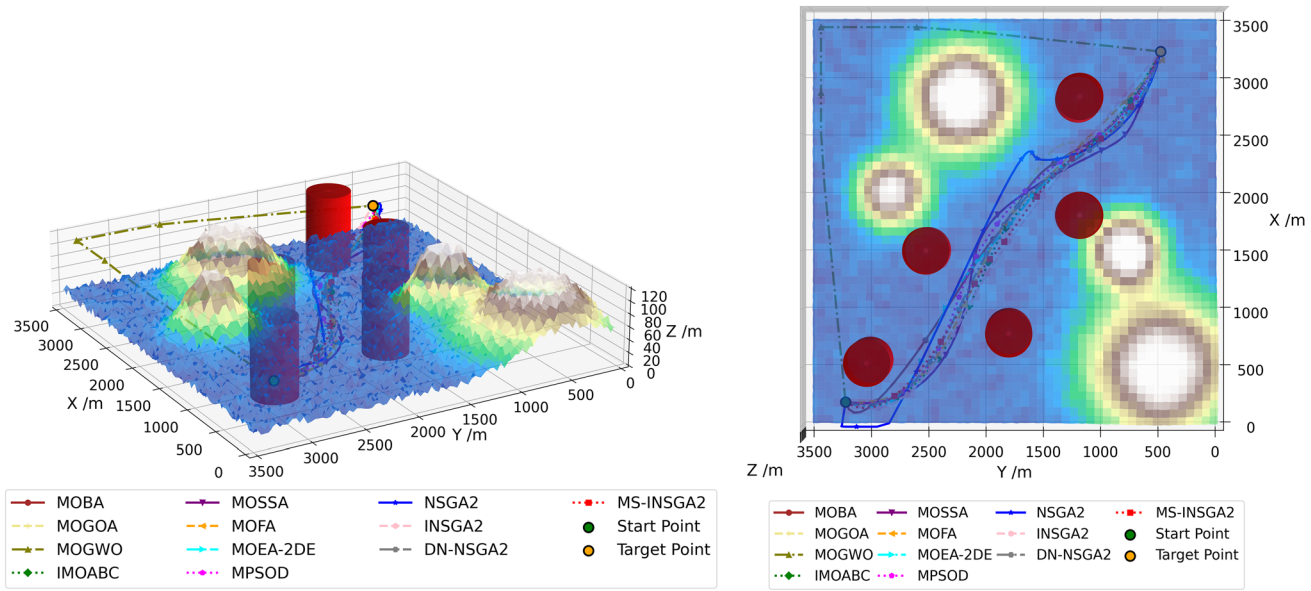
Experiment	Algorithm	Best Fitness	Average Fitness	Path length/m	Collision risk
Scene task Ⅰ	NSGA2	2.8260E+03	4.8674E+03	5.2261E+03	7.0996E+02
	A-INSGA2	2.5202E+03	4.7252E+03	4.7979E+03	4.0374E+02
	G-INSGA2	3.3031E+03	4.4439E+03	6.5751E+03	5.1534E+01
	L-INSGA2	2.1424E+03	2.5518E+03	4.2569E+03	2.1894E+01
	I-INSGA2	2.2277E+03	4.4351E+03	4.4187E+03	6.1235E+01
	MS-INSGA2	2.1317E+03	2.3677E+03	4.2495E+03	1.9767E+01

0.80%, 3.61%, 5.44%, 0.30%, 1.95%, 12.42%, 1.35%, and 0.62%, respectively, resulting in an overall average improvement of 6.91%.

To further validate the robustness of the algorithms in the Comprehensive I scenario, all evaluated algorithms were executed independently for 20 runs. The statistical results regarding the coefficient of variation (CV) of their fitness values are summarized in Table 15. The empirical data indicate that the mean fitness, standard deviation, and CV of the MS-INSGA2 algorithm are maintained at the lowest levels among all compared algorithms, with its CV being a mere 0.20%. Compared with the other 11 algorithms, MS-INSGA2 exhibits minimal optimization fluctuation and optimal stability. This thoroughly substantiates the algorithm's exceptional performance consistency, high reliability, and profound robustness in complex environments.

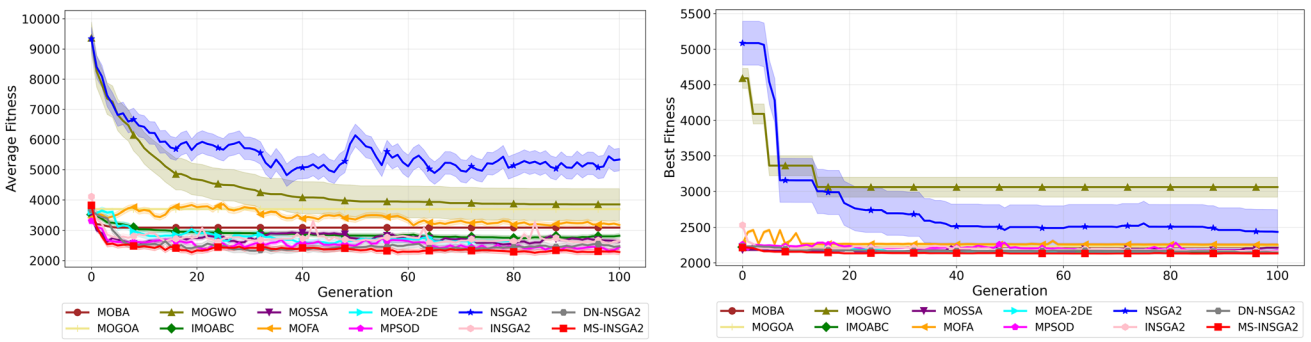
5.5.2 Comprehensive Ⅱ

According to the Scene task Ⅱ task arrangement in Table 12, the Comprehensive Ⅱ experiment is carried

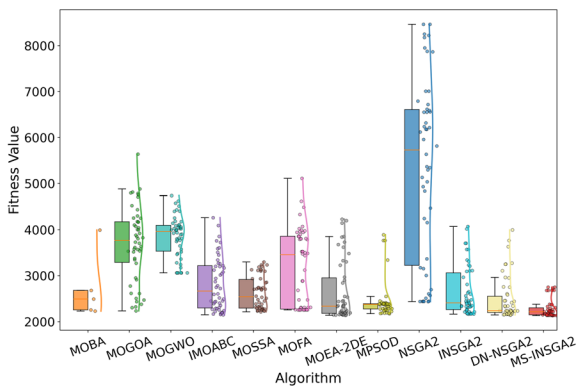


(a) First-person perspective (b) Top view perspective

Fig. 14 Task Ⅱ: 3D environmental feature distribution (comprehensive Ⅰ)



(a) Average Fitness Convergence Plot (b) Best Fitness Convergence Plot



(c) Box and violin picture Distribution

Fig. 15 Task Ⅱ: data visualization chart (comprehensive Ⅰ)

Table 14 Data of conclusion of comprehensive Ⅰ

Experiment	Algorithm	Best Fitness	Average Fitness	Path length/m	Collision risk
Comprehensive Ⅰ	MOBA	2.2286E+03	3.0906E+03	4.2496E+03	3.4583E+02
	MOGOA	2.2328E+03	3.6932E+03	4.2813E+03	3.0699E+02
	MOGWO	3.0595E+03	3.8492E+03	6.1189E+03	0.0000E+00
	IMOABC	2.1487E+03	2.8210E+03	4.2353E+03	1.0334E+02
	MOSSA	2.2113E+03	2.6294E+03	4.3825E+03	6.6924E+01
	MOFA	2.2541E+03	3.1784E+03	4.2736E+03	3.9101E+02
	MOEA-2DE	2.1380E+03	2.6476E+03	4.2426E+03	5.5597E+01
	MPSOD	2.1739E+03	2.4387E+03	4.2499E+03	1.6327E+02
	NSGA2	2.4339E+03	5.3363E+03	4.8603E+03	1.1909E+01
	INSGA2	2.1607E+03	2.6517E+03	4.2225E+03	1.6490E+02
	DN-NSGA2	2.1448E+03	2.4752E+03	4.2430E+03	7.7603E+01
	MS-INSGA2	2.1315E+03	2.2826E+03	4.2198E+03	7.2185E+01

out as shown in Fig. 16. The coordinates of the green starting point are set as (200,3200,60), the coordinates of the orange target end point are set as (3200,500,60), and the number of red cylindrical obstacles is set as 10. The middle track points of the path planning are set to 5, The locations are (250,2250,120), (550,3000,120), (1200,2000,120), (1200,3300,120), (1400,1200,120), (1700,2400,120), (2100,1300,120), (2300,2000,120), (2500,200,120), (3000,1000,120).

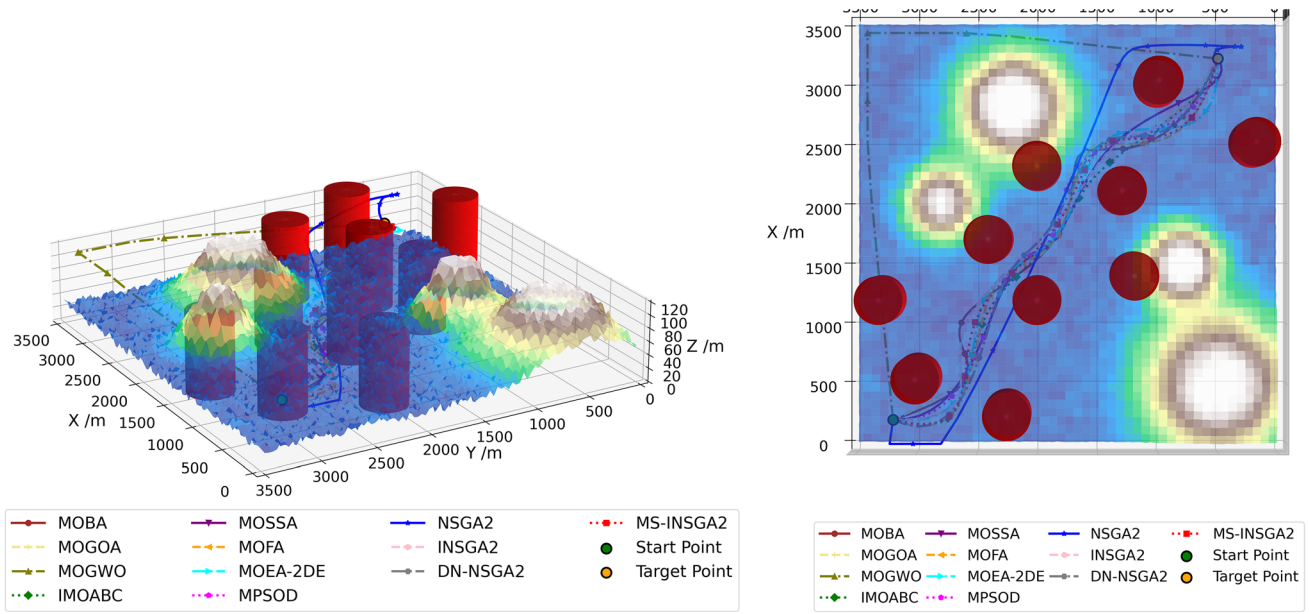
The experimental results of the Comprehensive II scenario are illustrated in Fig. 17. As observed from Figs. 17(a) and 17(b), the improved hybrid MS-INSGA2 algorithm demonstrates optimal performance in terms of both average and best convergence speeds when compared to the other evaluated algorithms. Furthermore, Fig. 17c corroborates the robustness of the hybrid MS-INSGA2 algorithm.

The performance comparison results between the improved hybrid strategy MS-INSGA2 algorithm and eleven algorithms—namely MOBA, MOGOA, MOGWO, IMOABC, MOSSA, MOFA, MOEA-2DE, MPSOD, NSGA2, INSGA2, and DN-NSGA2—in the Comprehensive II scenario are presented in Table 16. Regarding the Average Fitness, MS-INSGA2 achieves improvements of 33.94%, 34.87%, 33.91%, 17.04%, 46.96%, 30.48%, 18.47%, 16.49%, 44.89%, 33.40%, and 26.40%, respectively, compared to the counterpart algorithms, yielding an overall average improvement of 32.06%. In terms of the Best Fitness, it exhibits improvements of 17.66%, 16.95%, 24.63%, 12.95%, 17.34%, 10.53%, 2.33%, 7.37%, 24.69%, 11.79%, and 3.00%, respectively, resulting in an overall average improvement of 14.18%.

To further validate the robustness of the algorithms in the Comprehensive II scenario, all evaluated algorithms were executed independently for 20 runs. The statistical results regarding the coefficient of variation (CV) of their fitness values are summarized in Table 17. The empirical data indicate that the mean fitness, standard deviation, and CV of the MS-INSGA2 algorithm are maintained at the lowest levels among all compared algorithms, with

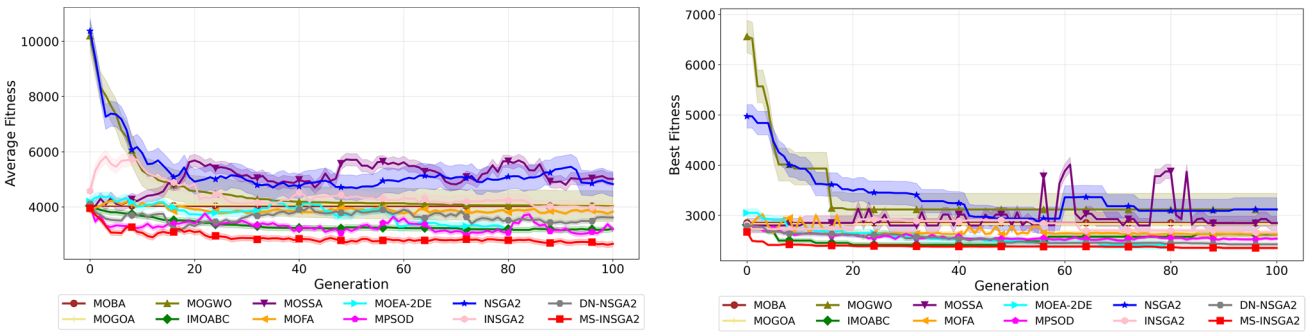
Table 15 Fitness coefficient of variation in comprehensive Ⅰ

Scene	Algorithm	Mean	Std	Fitness CV/%
Comprehensive Ⅰ	MOBA	2.6340E+03	2.4244E+02	9.20E+00
	MOGOA	2.6128E+03	7.0674E+01	2.70E+00
	MOGWO	3.0360E+03	1.0526E+02	3.47E+00
	IMOABC	2.1867E+03	1.9250E+01	8.80E-01
	MOSSA	2.3930E+03	1.0665E+02	4.46E+00
	MOFA	2.6169E+03	9.5698E+01	3.66E+00
	MOEA-2DE	2.1472E+03	7.6423E+00	3.60E-01
	MPSOD	2.3171E+03	1.1188E+02	4.83E+00
	NSGA2	2.7138E+03	2.3505E+02	8.66E+00
	INSGA2	2.1612E+03	1.3285E+01	6.10E-01
	DN-NSGA2	2.1737E+03	1.7918E+01	8.20E-01
	MS-INSGA2	2.1357E+03	4.2076E+00	2.00E-01

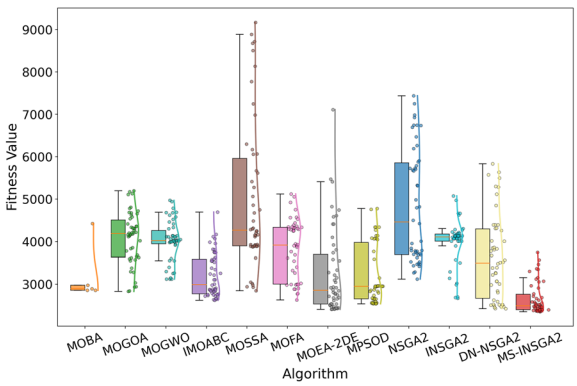


(a) First-person perspective (b) Top view perspective

Fig. 16 Task Ⅱ: 3D environmental feature distribution (comprehensive Ⅱ)



(a) Average Fitness Convergence Plot (b) Best Fitness Convergence Plot



(c) Box and violin picture Distribution

Fig. 17 Task Ⅱ: data visualization chart (comprehensive Ⅱ)

Table 16 Data of conclusion of comprehensive Ⅱ

Experiment	Algorithm	Best Fitness	Average Fitness	Path length/m	Collision risk
Comprehensive Ⅱ	MOBA	2.8540E+03	4.0285E+03	4.4880E+03	2.0332E+03
	MOGOA	2.8294E+03	4.0858E+03	4.3693E+03	2.1490E+03
	MOGWO	3.1180E+03	4.0263E+03	6.1191E+03	1.9491E+02
	IMOABC	2.6996E+03	3.2077E+03	4.3101E+03	1.5497E+03
	MOSSA	2.8427E+03	5.0167E+03	4.2747E+03	2.3512E+03
	MOFA	2.6266E+03	3.8280E+03	4.5460E+03	1.1788E+03
	MOEA-2DE	2.4059E+03	3.2641E+03	4.4999E+03	5.1967E+02
	MPSOD	2.5370E+03	3.1865E+03	4.3318E+03	1.2369E+03
	NSGA2	3.1201E+03	4.8288E+03	5.4869E+03	1.2548E+03
	INSGA2	2.6640E+03	3.9955E+03	4.3323E+03	1.6594E+03
	DN-NSGA2	2.4227E+03	3.6155E+03	4.4090E+03	7.2719E+02
	MS-INSGA2	2.3499E+03	2.6611E+03	4.4726E+03	3.7892E+02

its CV being a mere 0.37%. Compared with the other 11 algorithms, MS-INSGA2 exhibits minimal optimization fluctuation and optimal stability. This thoroughly substantiates the algorithm's exceptional performance consistency, high reliability, and profound robustness in complex environments.

5.5.3 Performance and robustness in scenarios I & II

A further quantitative evaluation of the two comprehensive scenarios, Comprehensive I and Comprehensive II, is presented in Table 18. The results demonstrate that, compared to the 11 baseline algorithms—namely MOBA, MOGOA, MOGWO, IMOABC, MOSSA, MOFA, MOEA-2DE, MPSOD, NSGA2, INSGA2, and DN-NSGA2—the proposed MS-INSGA2 algorithm achieves the global optimal performance in terms of both Generational Distance (GD) and Hypervolume (HV) evaluation metrics. The bold values in the table indicate the optimal results among all evaluated algorithms. Furthermore, the symbols "+", "−", and "≈" indicate that the performance of the proposed algorithm is significantly superior to, significantly inferior to, or statistically equivalent to that of the comparison algorithms, respectively, according to the Wilcoxon rank-sum test at a 0.05 significance level.

To further validate the actual robustness of the algorithms, this study introduces the Wilcoxon signed-rank test(Zheng et al. 2017) and effect size analysis (with the significance level set at 0.05). Differences between two sets of data are considered statistically significant when $p < 0.05$. To intuitively present the test results, statistical significance (Remarkable) is denoted as R, while non-significance (Not significant) is denoted as N. Regarding Cohen's d effect size, $d \geq 0.8$ (Great effect) is denoted as G, $0.5 \leq d < 0.8$ (Moderate effect) is denoted as M, and $d < 0.5$ (Minor effect) is denoted as S. Similarly, for the effect size r, $r \geq 0.5$, $0.3 \leq r < 0.5$, and $r < 0.3$ correspond to great, moderate, and minor effects, respectively, which are also denoted as G, M, and S accordingly. The detailed results of the Wilcoxon test and statistical effect size analysis for each algorithm under various evaluation metrics are presented in Table 19.

Table 17 Fitness coefficient of variation in comprehensive II

Scene	Algorithm	Mean	Std	Fitness CV/%
Comprehensive Ⅱ	MOBA	2.7839E+03	2.6249E+02	9.43E+00
	MOGOA	2.6520E+03	5.5508E+01	2.09E+00
	MOGWO	3.1180E+03	1.6684E+01	5.40E-01
	IMOABC	2.4398E+03	3.1614E+01	1.30E+00
	MOSSA	2.5378E+03	9.8668E+01	3.89E+00
	MOFA	2.6865E+03	1.4918E+02	5.55E+00
	MOEA-2DE	2.3962E+03	1.2014E+01	5.00E-01
	MPSOD	2.5791E+03	1.2416E+02	4.81E+00
	NSGA2	3.1153E+03	2.3683E+02	7.60E+00
	INSGA2	2.5979E+03	6.0468E+01	2.33E+00
	DN-NSGA2	2.4239E+03	2.9025E+01	1.20E+00
	MS-INSGA2	2.3562E+03	8.7362E+00	3.70E-01

Table 18 GD and HV metrics comparison: MS-INSGA2 vs other algorithms

Algorithm	Comprehensive Ⅰ		Comprehensive Ⅱ	
	GD	HV	GD	HV
MOBA	4.00E−03(−)	9.8460E+09(-)	1.50E−02(-)	2.3833E+10(-)
MOGOA	1.11E−02(-)	9.9155E+09(-)	1.11E−02(-)	2.4073E+10(-)
MOGWO	6.00E−03(-)	8.1284E+09(-)	2.43E−02(-)	2.0187E+10(-)
IMOABC	2.80E−03(-)	1.0061E+10(-)	4.4E−03(-)	2.4858E+10(-)
MOSSA	3.40E−03(-)	1.0121E+10(-)	1.10E−01(-)	2.3899E+10(-)
MOFA	7.10E−03(-)	9.8158E+09(-)	1.06E−02(-)	2.4779E+10(-)
MOEA-2DE	7.80E−03(-)	1.0121E+10(-)	5.50E−03(-)	2.5936E+10(-)
MPSOD	3.00E−03(-)	9.9936E+09(-)	7.00E−03(-)	2.5265E+10(-)
NSGA2	7.92E+01(-)	9.6413E+09(-)	2.15E+01(-)	2.4323E+10(-)
INSGA2	2.60E−03(-)	1.0027E+10(-)	6.00E−03(-)	2.4870E+10(-)
DN-NSGA2	2.10E−03(-)	1.0126E+10(-)	9.70E−03(-)	2.5948E+10(-)
MS-INSGA2	**1.80E−03**	**1.0129E+10**	**4.20E−03**	**2.5978E+10**
W±/≈	0/11/0	0/11/0	0/11/0	0/11/0

6 Sensitivity analysis

Having comprehensively verified the overall performance superiority of the MS-INSGA2 algorithm across various benchmark tests and trajectory planning tasks in the previous sections, this section further conducts a sensitivity analysis on the core hyperparameters of the algorithm. The proposed framework introduces multiple key control parameters, including dynamic guidance strength, perturbation range, mutation amplitude, and adaptive crossover and mutation boundaries. This section systematically evaluates the perturbing effects of these parameter variations on the final optimization performance. The objective is not only to determine the optimal parameter configuration suitable for multi-objective trajectory planning tasks but, more importantly, to verify the performance stability of the algorithm under different parameter settings. This analysis strongly supports the robustness and practical reliability of the proposed algorithm framework, proving that its excellent performance stems from the rationality of the mechanism design rather than the contingency of specific parameters.

6.1 Sensitivity analysis of dynamic guidance intensity coefficient

The dynamic guidance intensity coefficient s in Eq. 5 employs a linear decay mechanism. During the initial stage of the algorithm, $s = 0.5$, providing robust convergence guidance to steer the population toward the optimal region. As the iterations proceed, it linearly decreases to 0.1, substantially reducing the guidance intensity at the terminal stage, thereby effectively preventing the algorithm from falling into local optima. To verify the effectiveness of the value range for the dynamic intensity coefficient, sensitivity experiments were conducted by assigning different values, as illustrated in Fig. 18 and Table 20.

Based on the data presented in Fig. 18 and Table 20, it can be concluded that when s linearly decreases from 0.5 to 0.1 as the iterations proceed, both the Final best fitness and Mean best fitness achieve optimal results in multi-objective trajectory planning.

6.2 Sensitivity analysis of perturbation range

The selection of the perturbation range $U(-200, 200)$ in Eq. 6 is strictly determined based on the scale of the problem space and the constraint conditions. To verify the effectiveness of the chosen perturbation range, sensitivity experiments were conducted by adopting $U(-50, 50)$, $U(-100, 100)$, $U(-150, 150)$, and $U(-250, 250)$, respectively, as illustrated in Fig. 19 and Table 21.

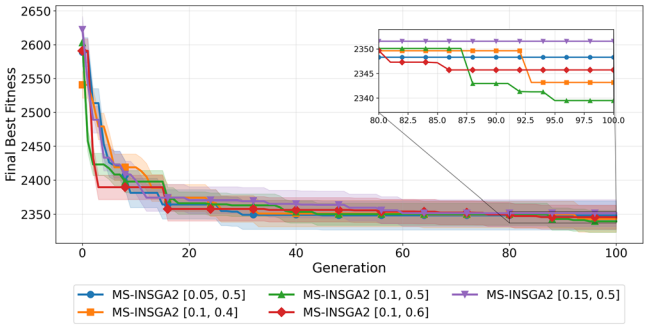
Based on the data presented in Fig. 19 and Table 21, it can be concluded that when the perturbation range is set to $U(-200, 200)$, both the Final best fitness and Mean best fitness achieve optimal results in multi-objective trajectory planning.

6.3 Sensitivity analysis of mutation amplitude parameter

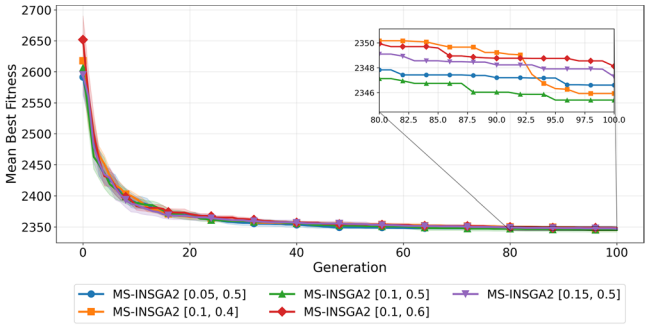
In the aforementioned Eq. 9, the initial and final amplitudes of the constant mutation probability amplitude range

Table 19 Wilcoxon test data (comprehensive Ⅰ & Ⅱ)

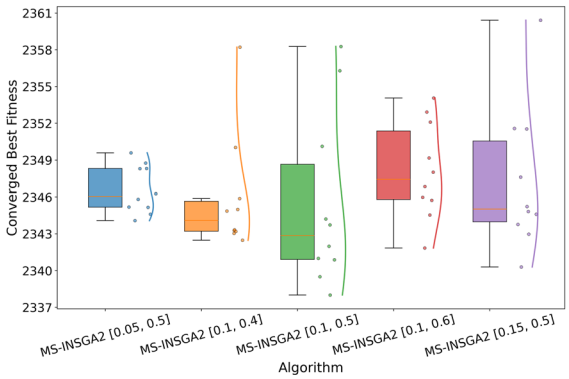
Indicators	Algorithm	Comprehensive Ⅰ			Comprehensive Ⅱ		
		p	*d*	*r*	*p*	*d*	*r*
Best Fitness	MOBA	1.91E-06 (R)	−2.8326(G)	−0.8765(G)	1.91E-06 (R)	−2.2448(G)	−0.876(G)
	MOGOA	1.91E-06 (R)	−9.2883(G)	−0.8765(G)	1.91E-06 (R)	−7.2564(G)	−0.8765(G)
	MOGWO	1.91E-06 (R)	−11.7805(G)	−0.8765(G)	1.91E-06 (R)	−119.8397(G)	−0.8765(G)
	IMOABC	1.91E-06 (R)	−3.5682(G)	−0.8765(G)	1.91E-06 (R)	−3.5163(G)	−0.8765 (G)
	MOSSA	1.91E-06 (R)	−3.3232(G)	−0.8765(G)	1.91E-06 (R)	−2.5281(G)	−0.8765(G)
	MOFA	1.91E-06 (R)	−6.9246(G)	−0.8765(G)	1.91E-06 (R)	−3.0473(G)	−0.8765 (G)
	MOEA-2DE	3.81E-06 (R)	−1.8188(G)	−0.8682(G)	1.91E-06 (R)	−3.7191(G)	−0.8765(G)
	MPSOD	1.91E-06 (R)	−2.2333(G)	−0.8765(G)	1.91E-06 (R)	−2.4688(G)	−0.8765(G)
	NSGA2	1.91E-06 (R)	−3.3896(G)	−0.8765(G)	1.91E-06 (R)	−4.4152(G)	−0.8765 (G)
	INSGA2	1.91E-06 (R)	−2.5244(G)	−0.8765(G)	1.91E-06 (R)	−5.4529(G)	−0.8765 (G)
	DN-NSGA2	1.91E-06 (R)	−2.8481(G)	−0.8765(G)	1.91E-06 (R)	−3.0802(G)	−0.8765 (G)
Average Fitness	MOBA	1.91E-06 (R)	−8.9744(G)	−0.8765(G)	1.91E-06 (R)	−5.3833(G)	−0.8765 (G)
	MOGOA	1.91E-06 (R)	−16.2320(G)	−0.8765(G)	1.91E-06 (R)	−9.5474(G)	−0.8765(G)
	MOGWO	1.91E-06 (R)	−10.4807(G)	−0.8765(G)	1.91E-06 (R)	−9.7170(G)	−0.8765 (G)
	IMOABC	1.91E-06 (R)	−5.0420(G)	−0.8765(G)	1.91E-06 (R)	−2.3786(G)	−0.8765 (G)
	MOSSA	1.91E-06 (R)	−2.1384(G)	−0.8765(G)	1.91E-06 (R)	−1.8108(G)	−0.8765 (G)
	MOFA	1.91E-06 (R)	−10.7838(G)	−0.8765(G)	1.91E-06 (R)	−6.9827(G)	−0.8765 (G)
	MOEA-2DE	3.62E-05(R)	−1.8642(G)	−0.8181(G)	1.91E-06 (R)	−3.1334(G)	−0.8765 (G)
	MPSOD	1.91E-06 (R)	−3.3998(G)	−0.8765(G)	1.91E-06 (R)	−2.9312(G)	−0.8765 (G)
	NSGA2	1.91E-06 (R)	−1.6655(G)	−0.8765(G)	1.91E-06 (R)	−4.8305(G)	−0.8765(G)
	INSGA2	9.54E-06 (R)	−2.0499(G)	−0.8515(G)	1.91E-06 (R)	−5.6825(G)	−0.8765 (G)
	DN-NSGA2	2.67E-05 (R)	−2.2144(G)	−0.8264(G)	3.81E-06 (R)	−2.3385(G)	−0.8682 (G)
Path length	MOBA	2.31E-01 (N)	0.3179(S)	−0.2755(S)	1.77E-01 (N)	−0.5023(M)	−0.3089 (M)
	MOGOA	3.28E-02 (R)	−0.8241(G)	−0.4758(M)	8.69E-01 (N)	0.0026(S)	−0.0417(S)
	MOGWO	1.91E-06 (R)	−9.3170(G)	−0.8765(G)	1.91E-06 (R)	−64.0226(G)	−0.8765 (G)
	IMOABC	1.335E-05(R)	1.8189(G)	−0.8431(G)	1.34E-05 (R)	1.7906(G)	−0.8431 (G)
	MOSSA	2.96E-02 (R)	−0.7953(M)	−0.4842(M)	2.02E-01 (N)	0.3332(S)	−0.2922 (S)
	MOFA	1.33E-01 (N)	−0.6629(M)	−0.3423(M)	1.33E-01 (N)	−0.5053(M)	−0.3423 (M)
	MOEA-2DE	3.49E-01 (N)	−0.3433(S)	−0.2170(S)	2.71E-03 (R)	1.0894(G)	−0.6428(G)
	MPSOD	1.77E-01 (N)	0.3976(S)	−0.3089(M)	1.72E-02 (R)	0.8688(G)	−0.5259 (G)
	NSGA2	1.34E-05 (R)	−1.8307(G)	−0.8431(G)	1.91E-06 (R)	−2.5475(G)	−0.8765 (G)
	INSGA2	1.05E-04 (R)	1.6878(G)	−0.7847(G)	6.74E-01 (N)	0.0723(S)	−0.1002 (S)
	DN-NSGA2	3.15E-03 (R)	−0.9967(G)	−0.6344(G)	1.33E-01 (N)	0.4903(S)	−0.3423 (M)
Collision risk	MOBA	1.91E-06 (R)	−2.7160(G)	−0.8765(G)	1.91E-06 (R)	−1.7588(G)	−0.8765(G)
	MOGOA	1.91E-06 (R)	−7.7052(G)	−0.8765(G)	1.91E-06 (R)	−4.8636(G)	−0.8765(G)
	MOGWO	6.37E-02 (N)	−0.0764(S)	−0.4174(M)	1.91E-06 (R)	7.0121(G)	−0.8765(G)
	IMOABC	1.91E-06 (R)	−3.9558(G)	−0.8765(G)	1.91E-06 (R)	−3.5133(G)	−0.8765(G)
	MOSSA	1.91E-06 (R)	−2.4169(G)	−0.8765(G)	1.91E-06 (R)	−2.1295(G)	−0.8765(G)
	MOFA	1.91E-06 (R)	−7.0002(G)	−0.8765(G)	1.91E-06 (R)	−2.2671(G)	−0.8765(G)
	MOEA-2DE	8.26E-02 (N)	−0.5990(M)	−0.3923(M)	1.91E-06 (R)	−2.8380(G)	−0.8765(G)
	MPSOD	1.91E-06 (R)	−2.1099(G)	−0.8765(G)	1.91E-06 (R)	−2.3689(G)	−0.8765(G)
	NSGA2	1.05E-04 (R)	−1.2779(G)	−0.7847(G)	5.86E-04 (R)	−1.3721(G)	−0.7179(G)
	INSGA2	3.81E-06 (R)	−2.4567(G)	−0.8682(G)	1.91E-06 (R)	−3.6218(G)	−0.8765(G)
	DN-NSGA2	1.21E-02 (R)	−0.8759(G)	−0.5510(G)	1.91E-06 (R)	−2.5981(G)	−0.8765(G)



(a) Convergence curves of final best fitness



(b) Convergence curves of mean best fitness



(c) Box plot of converged fitness distribution

Fig. 18 Sensitivity and performance analysis of dynamic guidance strength

Table 20 Experimental conclusions on guidance intensity coefficient range

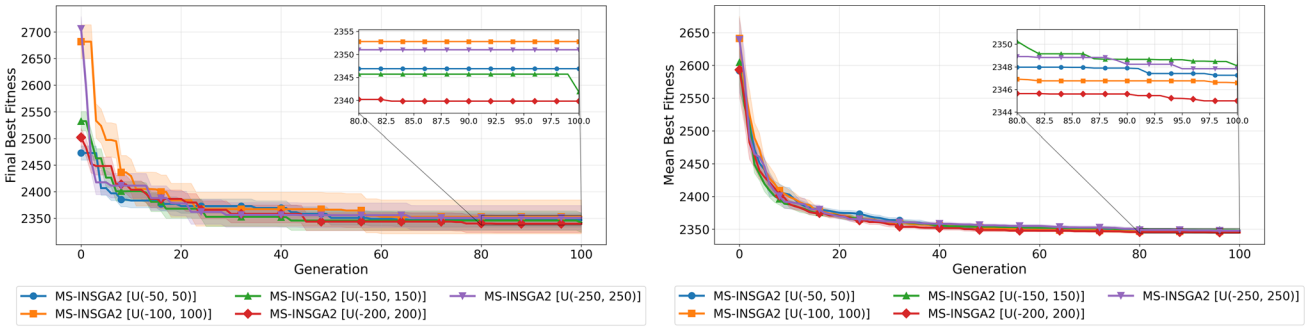
Guidance strength range	Final best fitness	Mean best fitness	Path length/m	Collision risk
[0.05, 0.5]	2.3483E+03	2.3466E+03	4.4374E+03	4.3203E+02
[0.1, 0.4]	2.3432E+03	2.3459E+03	4.4247E+03	4.3608E+02
[0.1, 0.5]	2.3395E+03	2.3454E+03	4.4586E+03	3.6739E+02
[0.1, 0.6]	2.3457E+03	2.3481E+03	4.4198E+03	4.5279E+02
[0.15, 0.5]	2.3516E+03	2.3473E+03	4.4042E+03	4.9821E+02

are set to 80 m and 20 m, respectively. To verify the effectiveness of these amplitude parameters, sensitivity comparative experiments were conducted by setting the initial and final amplitudes to [60, 20], [70, 20], [90, 20], [80, 10], and [80, 30], respectively, as illustrated in Fig. 20 and Table 22.

Based on the data presented in Table 22 derived from the aforementioned sensitivity experiments, it can be concluded that when the initial and final amplitudes of the mutation probability amplitude range are set to 80 m and 20 m, respectively, both the Final best fitness and Mean best fitness achieve optimal results in multi-objective trajectory planning.

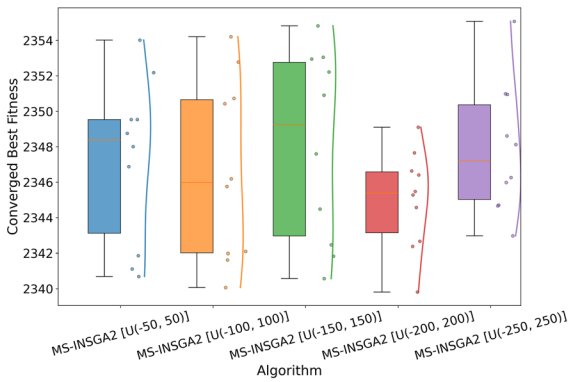
6.4 Sensitivity analysis of adaptive crossover and mutation

To verify the effectiveness of the adaptive crossover and mutation operators, sensitivity experiments were conducted using the configuration proposed in this study (Proposed Baseline: 0.75, 0.65, 0.25, 0.3) as the benchmark. Firstly, under the condition of a fixed mutation rate, the impacts of a higher crossover rate (High: 0.85, 0.75, 0.25, 0.3) and a lower crossover rate (Low: 0.65, 0.55, 0.25, 0.3) were evaluated, respectively. Secondly, under the condition of a fixed crossover rate, the



(a) Convergence curves of final best fitness

(b) Convergence curves of mean best fitness



(c) Box plot of converged fitness distribution

Fig. 19 Sensitivity analysis and performance comparison of the perturbation range

Table 21 Experimental results of perturbation range

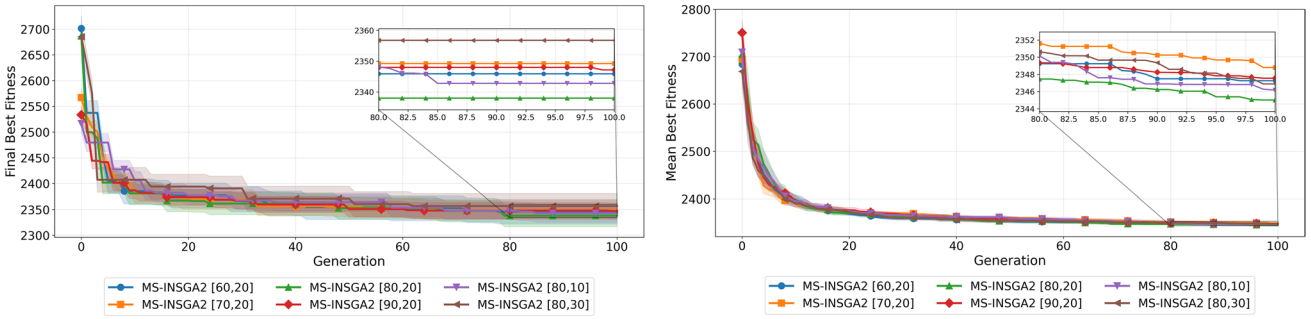
Perturbation range	Final best fitness	Mean best fitness	Path length/m	Collision risk
U(−50, 50)	2346.8793	2347.2590	4430.7714	438.3004
U(−100, 100)	2352.7834	2346.5859	4410.8857	491.1222
U(−150, 150)	2341.8395	2348.0908	4394.1722	482.5006
U(−200, 200)	2339.8152	2344.9958	4422.8501	427.9547
U(−250, 250)	2350.9924	2347.8336	4397.1123	508.1091

perturbations to the algorithmic performance caused by a higher mutation rate (High: 0.75, 0.65, 0.35, 0.4) and a lower mutation rate (Low: 0.75, 0.65, 0.15, 0.25) were further examined. Furthermore, a set of fixed parameter configurations (Fixed Operators: 0.65, 0.65, 0.15, 0.15) was introduced as a control group devoid of any adaptive mechanism. The detailed comparative results of the sensitivity experiments are illustrated in Fig. 21.

Based on the comparative experimental data for the adaptive operator parameters presented in Table 23, it can be concluded that when the initial value of the adaptive crossover probability $\psi_{c,init} = 0.75$, the final value $\psi_{c,final} = 0.65$, the base value of the adaptive mutation probability $\psi_{m,bas} = 0.25$, and the maximum adaptive mutation probability is set to $\psi_{m,max=0.3}$,, both the Final best fitness and Mean best fitness of the proposed algorithm achieve optimal results in the multi-objective trajectory planning task.

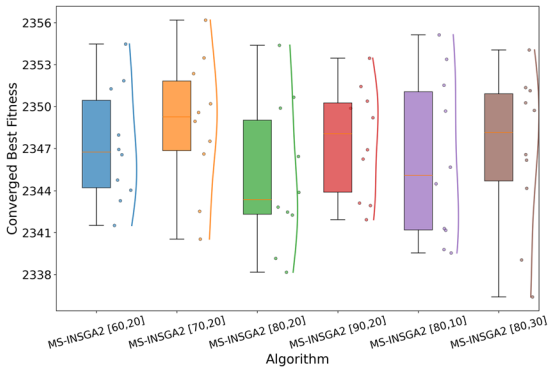
6.5 Sensitivity analysis of cooling coefficient

The sensitivity comparative experiments for the cooling coefficient α are illustrated in Fig. 22. The comparative values for multiple sets of the cooling coefficient α are set



(a) Convergence curves of final best fitness

(b) Convergence curves of mean best fitness



(c) Box plot of converged fitness distribution

Fig. 20 Sensitivity analysis and performance comparison of key parameter

Table 22 Experimental conclusions on mutation probability amplitude range

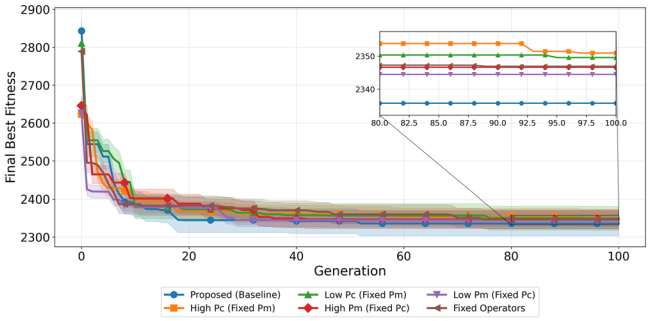
Initial amplitude/m	Final amplitude/m	Final best fitness	Mean best fitness	Path length/m	Collision risk
60	20	2345.8959	2347.2630	4455.5610	446.7643
70	20	2349.2301	2348.7975	4435.2492	513.7226
80	20	2337.9968	2345.0157	4428.4581	412.5467
90	20	2347.1483	2347.5514	4411.6887	439.2484
80	10	2342.8055	2346.1676	4368.1424	422.3797
80	30	2356.7627	2346.8890	4443.3941	485.0927

to 0.8, 0.95, and 0.99, respectively, and the resulting data are presented in Table 24.

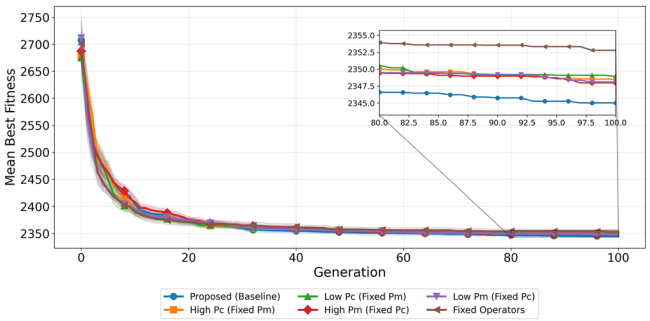
Based on the aforementioned sensitivity experiments, it can be concluded that when the cooling coefficient is set to 0.9, both the Final best fitness and Mean best fitness achieve optimal results in multi-objective trajectory planning.

6.6 Sensitivity analysis of diversity threshold

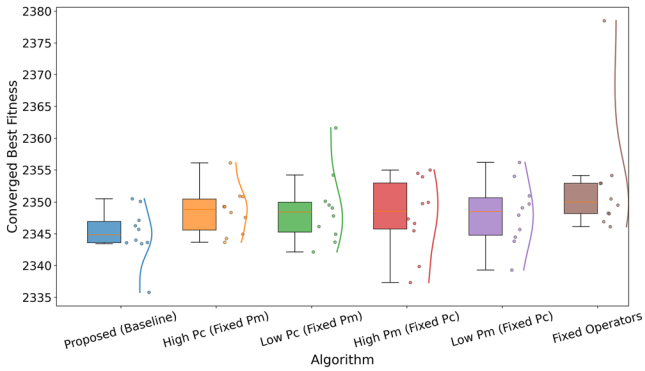
The threshold for triggering Eq. 6 is set to a fixed parameter, namely $D = 0.15$. When $D < 0.15$, the algorithm determines that the population has fallen into aggregation stagnation. Sensitivity comparative experiments for the threshold were conducted as illustrated in Fig. 23, with the comparative



(a) Convergence curves of final best fitness



(b) Convergence curves of mean best fitness



(c) Box plot of converged fitness distribution

Fig. 21 Sensitivity analysis of adaptive crossover and mutation operators

Table 23 Experimental conclusions on the sensitivity of the adaptive operators

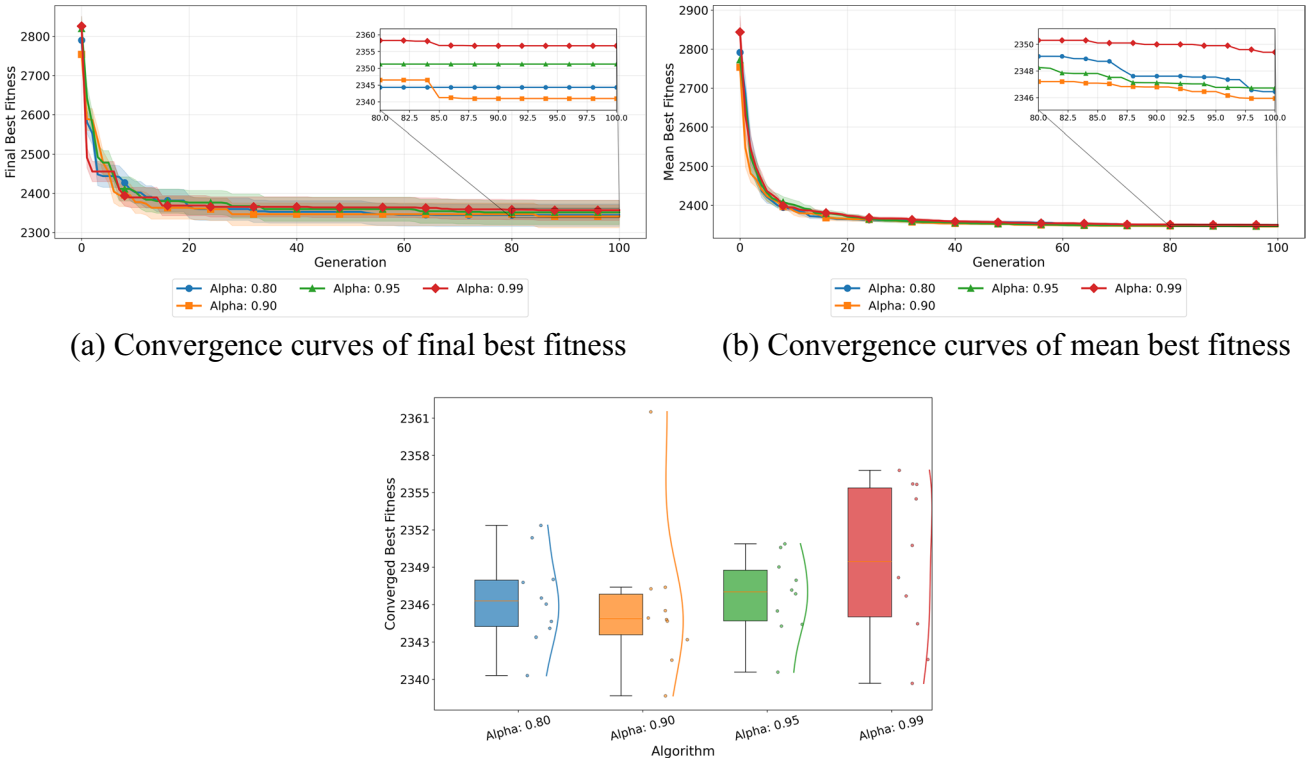
Adaptive operator parameters	Final best fitness	Mean best fitness	Path length/m	Collision risk
Proposed (Baseline)	2335.7896	2345.0188	4445.5009	376.7830
High Pc (Fixed Pm)	2350.9247	2348.5219	4477.2922	374.2472
Low Pc (Fixed Pm)	2349.5353	2348.9260	4458.4047	401.0955
High Pm (Fixed Pc)	2346.6328	2347.9727	4414.2029	465.0923
Low Pm (Fixed Pc)	2344.4657	2348.1160	4424.8047	440.1982
Fixed Operators	2346.9171	2352.7842	4387.4363	510.6519

parameters set to 0.05, 0.10, 0.20, and 0.25, respectively. The resulting data are presented in Table 25.

The results of the aforementioned sensitivity analysis experiments demonstrate that when the diversity threshold is set to D = 0.15, the algorithm achieves the optimal Final best fitness and Mean best fitness in the multi-objective trajectory planning task. This result fully substantiates the rationality and scientific validity of the empirical parameter settings in this study.

7 Conclusion

To address the unmanned aerial vehicle (UAV) trajectory planning problem in complex three-dimensional (3D) environments, this paper proposes a hybrid multi-objective optimization algorithm, named MS-INSGA2, which integrates multiple improvement strategies. The core contributions of this study lie in optimizing the quality of initial solutions through a linear interpolation mechanism, enhancing



(a) Convergence curves of final best fitness

(b) Convergence curves of mean best fitness

(c) Box plot of converged fitness distribution

Fig. 22 Sensitivity analysis and performance comparison of the cooling coefficient

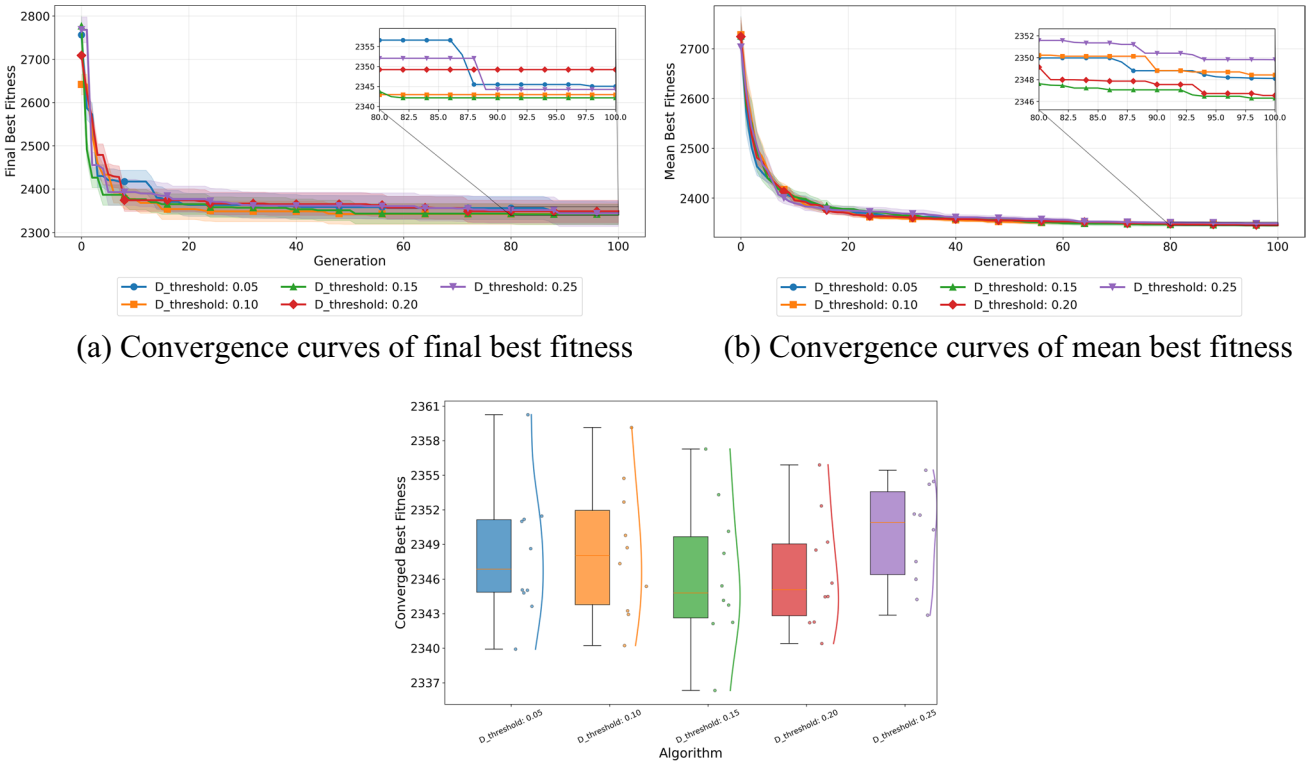
Table 24 Experimental conclusions on the cooling coefficient

Alpha	Final best fitness	Mean best fitness	Path length/m	Collision risk
0.80	2344.3214	2346.4467	4437.8566	441.4504
0.90	2340.9980	2345.9436	4437.8362	419.1173
0.95	2351.2526	2346.7194	4414.1779	457.2244
0.99	2356.7130	2349.3976	4434.8059	444.4869

global exploration efficiency by utilizing dynamic guidance alongside adaptive crossover and mutation strategies, and strengthening solution precision in combination with a local search mechanism. This hybrid architecture effectively breaks the bottleneck of balancing convergence and population diversity in traditional multi-objective optimization, significantly improving the algorithm's capability to escape local optima.

Extensive experimental analyses profoundly validate the superiority of the proposed algorithm. Its performance on the standard ZDT, DTLZ, and WFG benchmark test suites substantiates that MS-INSGA2 possesses outstanding mathematical optimization capabilities and underlying robustness. In the subsequent comprehensive experiments of 3D spatial trajectory planning (Comprehensive I and II), ablation studies explicitly confirm the presence of a strong positive synergistic effect among the four improvement mechanisms. Compared to 11 state-of-the-art algorithms such as NSGA2, MOBA, and MOGOA, MS-INSGA2 comprehensively achieves the optimal Mean best fitness and Final best fitness without sacrificing computational stability. Specifically, the proposed algorithm is capable of planning shorter flight paths while significantly reducing the risk of spatial collisions, thereby generating smoother flight trajectories with higher safety.

Despite the exceptional performance demonstrated by MS-INSGA2, the current research still presents certain limitations. For instance, the current model is primarily constructed based on static threat environments, and the integration of multiple strategies increases the



(a) Convergence curves of final best fitness

(b) Convergence curves of mean best fitness

(c) Box plot of converged fitness distribution

Fig. 23 Sensitivity analysis and performance comparison of the diversity threshold

Table 25 Experimental conclusions on the diversity threshold parameter

Diversity threshold	Final best fitness	Mean best fitness	Path length/m	Collision risk
0.05	2345.0440	2348.1010	4405.0803	475.0018
0.10	2342.9408	2348.4187	4397.3197	480.9234
0.15	2342.1412	2346.3009	4435.5168	414.5964
0.20	2349.2022	2346.5501	4420.4942	463.1717
0.25	2344.2468	2349.8175	4401.8588	477.7126

computational complexity of the algorithm to some extent. Future research will focus on the following prospective directions: First, introducing dynamic threat models (e.g., sudden obstacles or moving targets) to evaluate the real-time replanning capability of the algorithm in highly dynamic environments. Second, incorporating real-world, real-time meteorological information (such as wind fields and extreme weather) and UAV energy consumption models into the objective constraints. Finally, extending the framework to cooperative planning for multi-UAV swarms to investigate multi-aircraft collision avoidance and task allocation mechanisms in dense airspace. These future studies will not only further enhance the applicability of the algorithm in complex real-world scenarios but also provide more solid theoretical and technical support for fields such as large-scale logistics distribution and emergency rescue in the context of the low-altitude economy.

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Author contribution Author Jian Deng designed the experiments, proposed the MS-INSGA2 algorithm and the hybrid strategy-improved UAV trajectory planning method, implemented the trajectory planning algorithms, performed the comprehensive experimental evaluations, and wrote the original draft. Author Honghai Zhang supervised the project, provided guidance and funding support, and reviewed and edited the paper. Author Dikun Feng assisted in designing the optimization strategies and conducting the statistical validation. Author Zeyu Liu assisted in the comprehensive literature review, as well as data organization and verification. Author Zihan Yu assisted in organizing the experimental data and results. All authors reviewed the manuscript.

Data availability All data generated or analyzed during this study are included in this published article and its supplementary information files.

Code availability The core source code required to reproduce the main algorithmic findings of this study, including the proposed MS-INSGA2 algorithm, its comparative evaluation against the classical NSGA-II, and the ablation experiments, is publicly available at: https://github.com/dj-dengjian/A-Hybrid-Multi-Objective-Optimization-Algorithm-for-UAV-Trajectory-Planning.

Declarations

Ethics Not applicable.

Competing interest The authors declare no competing interests.

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